

The processing of ambiguous words without context : Supplementary material

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Experimental material

1 LDT

1.1 Procedure

Participants were seated 70 cm in front of a screen. Figure 1 illustrates the sequence of stimuli presented to them, which included a fixation cross (200 ms), a prime word (500 ms), a blank screen (500 ms), and a target stimulus. The target was either a real French word or a non-word and remained on the screen until the participant responded.

Participants were instructed to ignore the first word and determine whether the second was a valid French word ("yes") or not ("no"), responding as quickly as possible while minimizing errors. They were required to keep both hands on two designated computer keys, each marked with a colored sticker labeled O (yes) or N (no). The "O" response key was always assigned to the participant's dominant hand.

Before the main task, each participant completed 14 training trials to familiarize themselves with the sequence, instructions, and response keys. During training, they received feedback, and the instructions were clarified if needed. No feedback was provided during the experimental trials. Each participant completed 168 test trials, with word presentation randomized by the program.

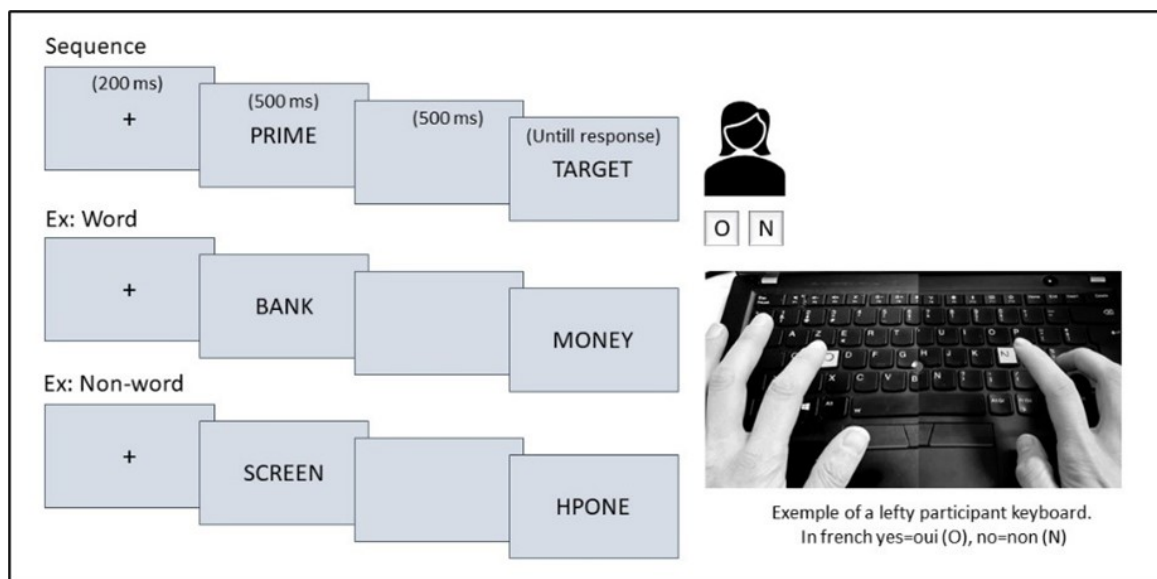


FIGURE 1 – LDT Sequence

1.2 Material

1.2.1 Corpus of ambiguous words

The ambiguous words and their dominant/subordinate meanings were selected from the corpus normed by previous authors (THÉROUANNE et DENHIÈRE, 2004).

THÉROUANNE et DENHIÈRE (2004) selected 162 ambiguous words from the dictionary (e.g., carrière [CAREER]), avoiding Anglicisms and words with overly technical meanings. A panel of 200 students then underwent a free association task based on this corpus. Two independent

researchers subsequently reclassified the words resulting from these free associations, creating categories corresponding to the meanings (semantically different) of ambiguous words REZLESCU et al., 2020.

The pairs determined by THÉROUANNE et DENHIÈRE (2004) were established with a population of young adults. Therefore, for the present research, we added a phase of judgment of polarity by older participants. We asked 51 participants (mean age : 71.38 $\pm$ 4.75) to judge the polarity. The participants were asked to evaluate pairs of words composed of an ambiguous word with either its dominant or subordinate meaning, or an unrelated word. They were asked to rate the strength of the link on a Likert scale from 1 to 7 (with 7 indicating the strongest relation). Only the triads ambiguous word-dominant meaning-subordinate meaning for which the polarity was significantly different were retained.

The Table shows overall differences between senses (Friedman tests). Significant results indicate that participants did not treat all senses equally. Wilcoxon pairwise comparisons reveal that one meaning (dominant) is usually preferred over others. Some pairs showed no significant difference, suggesting ambiguity or coexistence between senses. This pairs were not retained in the material. At the end, we selected 30 ambiguous words to compose our ambiguous conditions, also considering the balance of frequency, concreteness, and number of orthographic neighbors.

THÉROUANNE et DENHIÈRE (2004) stress that the difference between homonymy and polysemy can significantly influence experimental results. Homonyms refer to distinct lexical entries that share the same form but have unrelated meanings (e.g., bark as tree covering vs. dog sound), whereas polysemous words correspond to semantically related senses within a single lexical entry. THÉROUANNE et DENHIÈRE (2004) note that many studies use polysemous and homonymous words interchangeably, which is likely to affect experimental results, and they underline that the presence or absence of semantic relatedness between meanings cannot be neglected in cognitive research on ambiguous words. Following this distinction, half of the ambiguous stimuli of the following protocol are homonyms and half are polysemous words.

Paires a		Résultats obtenus au test de Friedman		Résultats obtenus aux tests de Wilcoxon			
Homonyme	Sens dominant	X^2	p^*	paire vs paire		W	p^*
	Sens subordonné			paire vs paire			
	Sens non relié			paire vs paire			
Accès	Entrée	38.99	< .001	accès-entrée vs accès-douleur		210	<.001
	Douleur			accès-douleur vs accès-potage		45	.008
	Potage						
Affection	Amour	30.64	< .001	affection-amour vs affection- maladie		30	.104
	Maladie			affection-maladie vs affection- téléphone		181	<.001
	Téléphone						
Ampoule	Lumière	40.03	< .001	ampoule-lumière vs ampoule- cloque		42	.022
	Cloaque			ampoule-cloque vs ampoule-billet		253	<.001
	Billet						
Adresse	Maison	38.21	< .001	adresse-maison vs adresse-habileté		28.5	.959
	Habileté			adresse-habileté vs adresse-nuit		253	<.001
	Nuit						
Arête	Os	35.12	< .001	arête-os vs arête-carré		109.5	.005
	Carré			arête-carré vs arête-anglais		120	<.001
	Anglais						
Aube	Matin	32.40	< .001	aube-matin vs aube-prêtre		82	.011
	Prêtre			aube-prêtre vs aube-action		188.5	<.001
	Action						
Arc	Carquois	37.34	< .001	arc-carquois vs arc-triomphe		79	.003
	Triomphe			arc-triomphe vs arc-surligneur		179	<.001
	surligneur						
Artifice	Feu	25.73	< .001	artifice-feu vs artifice-superficiel		65	.182
	Superficiel			artifice-superficiel vs artifice-maquette		169	<.001
	Maquette						
Avocat	Juge	36.96	< .001	avocat-juge vs avocat-légume		86	.005
	Légume			avocat-légume vs avocat-moustique		210	<.001
	Moustique						
Bide	Ventre	30.48	< .001	bide-ventre vs bide-nul		121.5	.033
	Nul			bide-nul vs bide-fardeau		173	.002
	Fardeau						
Bidet	Lavabo	28.53	< .001	bidet-lavabo vs bidet-âne		143	.002
	Âne			bidet-âne vs bidet-danseur		66	.004
	Danseur						
Bretelle	Pantalon	39.69	< .001	bretelle-pantalon vs bretelle- autoroute		6	.766
	Autoroute			bretelle-autoroute vs bretelle- source		253	<.001
	Source						
Cafard	Insecte	35.53	< .001	cafard-insecte vs cafard-angoisse		66	.003
	Angoisse			cafard-angoisse vs cafard-antidote		208.5	<.001
	Antidote						
Calcul	Mathématique	38.33	< .001	calcul-mathématique vs calcul- rein		25.5	0.32
	Rein			calcul-rein vs calcul-rideau		253	<.001
	Rideau						
Canapé	Fauteuil	36.96	< .001	canapé-fauteuil vs canapé-apéritif		86	.005
	Apéritif			canapé-apéritif vs canapé-utopie		c	c
	Utopie						
Canne	à sucre	36.21	< .001	canne - à sucre vs canne - oiseau		67.5	.007
	Oiseau			canne-à sucre vs canne - fenêtre		198	<.001
	fenêtre						
Chenille	Papillon	34.82	< .001	chenille-papillon vs chenille-char		88	.003
	Char			chenille-char vs chenille-misère		188	<.001
	Misère						
Cliché	Photo	30.82	< .001	cliché-photo vs cliché-stéréotype		70	.28
	Stéréotype			cliché-stéréotype vs cliché- condition		209	<.001
	Condition						
Comté	Fromage	40.44	< .001	comté-fromage vs comté-région		17	.203
	Région			comté-région vs comté-flèche		253	<.001
	Flèche						
Couette	Couverture	29.79	< .001	couette-couverture vs couette- cheveux		47.5	.045
	Cheveux			couette-cheveux vs couette-vie		218	<.001
	Vie						
Entretien	Embauche	39.35	< .001	entretien-embauche vs entretien- ménage		22	.615
	Ménage			entretien-ménage vs entretien- plastique		253	<.001
	Plastique						
Faculté	Université	39.81	< .001	faculté-université vs faculté- capacité		15	.931
	Capacité			faculté-capacité vs faculté-métal		b	b

	Métal					
Filature	Détective	29.95	< .001	filature-détective vs filature-tissu	12.5	.134
	Tissu			filature-tissu vs filature-bisou	245.5	< .001
	bisou					
Fléau	Peste			fléau-peste vs fléau-arme	64.5	.006
	Arme	34.41	< .001	fléau-arme vs fléau-gourde	188	< .001
	Gourde					
Garrot	Sang			garrot-sang vs garrot-cheval	67.5	.026
	Cheval	37.12	< .001	garrot-cheval vs garrot-banc	231	< .001
	Banc					
Glace	Miroir			glace-miroir vs glace-vanille	36.5	.027
	Vanille	35.15	< .001	glace-miroir vs glace-accusé	173	.003
	Accusé			glace-miroir vs glace-accusé	79	.152
Légende	Histoire			légende-histoire vs légende- explication	105	.001
	Explication	37.08	< .001	légende-explication vs légende- loyer	136	< .001
	Loyer					
Motif	Raison	30.43	< .001	motif-raison vs motif-dessin	39.5	1
	Dessin			motif-dessin vs motif-copain	231	< .001
	Copain					
Navet	Carotte			navet-carotte vs navet-film	53	.011
	Film	38.91	< .001	navet-film vs navet-cravate		c
	Cravate					
Note	Musique	40.84	< .001	note-examen vs note-musique	0	.003
	Examen			note-musique vs note-cratère	253	< .001
	Cratère			note-examen vs note-cratère	253	< .001
Page	Livre			page-livre vs page-ménestrel	99	.004
	Ménestrel	32.14	< .001	page-ménestrel vs page-fourchette	120	< .001
	Fourchette					
Palais	Château	34.03	< .001	palais-château vs palais-bouche	59	.7
	Bouche			palais-bouche vs palais-chose	231	< .001
	Chose					
Parquet	Sol	40.78	< .001	parquet-sol vs parquet-justice	18.5	.106
	Justice			parquet-justice vs parquet-viande	253	< .001
	Viande					
Perception	Vision			perception-vision vs perception- impôt	6	.017
	Impôt	34.16	< .001	perception-impôt vs perception- épine	253	< .001
	Épine			perception-vision vs perception- épine	205	< .001
Pensée	Opinion	34.25	< .001	pensée-opinion vs pensée-fleur	29	.022
	Fleur			pensées-fleur vs pensée flèche	187	< .001
	Flèche					
Perche	Saut	30.19	< .001	perche-saut vs perche-poisson	12.5	.203
	Poisson			perche-poisson vs perche- ouverture	208	< .001
	Ouverture					
Pieu	Vampire	14.48	< .001	pieu-vampire vs pieu-lit	35.5	.029
	Lit			pieu-lit vs pieu-ligne	184.5	.003
	Ligne			pieu-vampire vs pieu-ligne	86	.145
Plongeur	Mer	38.99	< .001	plongeur-mer vs plongeur- vaisselle	52.5	.085
	Vaisselle			plongeur-vaisselle vs plongeur- enjeu	253	< .001
	Enjeu					
Pupille	Ceil	38.16	< .001	pupille-ciel vs pupille-nation	55	.005
	Nation			pupille-nation vs pupille-statue	190	< .001
	Statue					
Recette	Cuisine	36.20	< .001	recette-cuisine vs recette-argent	42	0.021
	Argent			recette-argent vs recette-politique	210	< .001
	Politique					
Réflexion	Pensée			réflexion-pensée vs réflexion- reflet	147	< .001
	Reflet	34.78	< .001	réflexion-reflet vs réflexion- poubelle	136	< .001
	Poubelle					
Siège	Chaise			siège-chaise vs siège-société	98	.001
	Société	28.97	< .001	siège-société vs siège-cadre	153	< .001
	Problème	34.53	< .001	solution-problème vs solution- chimie	158	.002
Solution	Chimie			solution-chimie vs solution- étagère	207.5	< .001
	Souffle			soupir-souffle vs soupir-silence	46.5	.727
	Silence	33.71	< .001	soupir-silence vs soupir-paysage	231	< .001
Soupir	Paysage					
	Football			stade-football vs stade-étape	187	< .001
	Étape	30.73	< .001	stade-étape vs stade-autre	128.5	.014
Stade*	Autre					
	Étagère					
Timbre	Poste	40.03	< .001	timbre-poste vs timbre-voix	42.5	.019
	Voix			timbre-voix vs timbre-pilote	253	< .001

	Pilote					
Trafic	Voiture	38.75	< .001	trafic-voiture vs trafic-drogue	22.5	1
	Drogue			trafic-drogue vs trafic-loutre	253	<.001
	Loutre					
Tuteur	Parent	36.03	< .001	tuteur-parent vs tuteur-plante	34.5	.749
	Plante			tuteur-plante vs tuteur-filtre	231	<.001
	Filtre					
Vase	Fleur			vase-fleur vs vase-boue	102	.002
	Boue	29.92	< .001	vase-boue vs vase-capsule	163.5	<.001
	Capsule					

1.2.2 Counterbalancement

Four versions of the task were created, each including 168 pairs of words. The complete corpus of words included 30 ambiguous words, paired with a semantic associate corresponding to their dominant (AD ; N=30) et subordinated (AS ; N=30) meaning, is provided in 3. Lexical Decision Task ; Pairs of related prime-target with monosemic prime). The dominant and subordinated target were reassigned as semantically associated targets (M ; N=60) to monosemic primes. All the words were then reassigned to compose the 120 non related pairs. Non-words, constructed by reversing letters from actual words, according to Yap et al.s (2013) recommendations, were taken from the corpus by Labalestra (2018). Every version therefore included 64 counterbalanced pairs (8 AD, 8 AS, 16 M, 32 NR), 20 fill-in non-counterbalanced pairs and 84 pairs with a non-word as target (equal for the four versions). See Table ?? for a representation of the counterbalancement between the conditions and the versions.

Every version of the task existed in a lefty- or righty- adapted version, in which the Yes answer was on the dominant side of the table.

In the counterbalanced conditions (AD, AS, M, NR) the target were balanced on the frequency (book and movie), the number of letters, the orthographic neighborhood and the concreteness, and the prime were balanced on the frequency (book and movie), the orthographic neighborhood and the concreteness.

PRIME		TARGET	
EXAMPLE VERSION 1			
AD (8 trials)	e.g. BANK	e.g. MONEY	Counterbalanced
AS (8 trials)	e.g. RIGHT	e.g. DIRECTION	Counterbalanced
M (16 trials)	e.g. PHONE	e.g. DIGIT	Counterbalanced
NR (32 trials)	e.g. GRASS	e.g. COMPUTER	Counterbalanced
Fill-in NR (20 trials)	e.g. SUGAR	e.g. PLANT	Not counterbalanced
Non-words (84 trials)	e.g. NOSE	e.g. PUTCIRE	Equal among versions
EXAMPLE VERSION 2			
AD (8 trials)	e.g. RIGHT	e.g. CORRECT	Counterbalanced
AS (8 trials)	e.g. BANK	e.g. RIVER	Counterbalanced
M (16 trials)	e.g. SAVING	e.g. MONEY	Counterbalanced
NR (32 trials)	e.g. PHONE	e.g. DIRECTION	Counterbalanced
Fill-in NR (20 trials)	e.g. PLANT	e.g. SUGAR	Not counterbalanced
Non-words (84 trials)	e.g. NOSE	e.g. PUTCIRE	Equal among versions

Note.

TABLE 1 – Exemple of stimuli and counterbalancing between versions

Each version includes 8 AD trials, 8 AS trials, 16 M trials, 32 NR trials, 20 Fill-in NR trials, and 84 Non-word trials. The first four categories were counterbalanced across four versions to ensure that each participant saw a different subset of stimuli. The Fill-in NR items were not counterbalanced, and the Non-word items were evenly distributed among versions. Participants were assigned versions in a rotating order (Participant 1 : Version 1 ; Participant 2 : Version 2 ; Participant 3 : Version 3 ; Participant 4 : Version 4 ; Participant 5 : Version 1 ; etc.).

1.2.3 SOA

The automatic process measured with SPE is linked to stimulus onset asynchrony (SOA), which is the time lapse between the appearance of the prime and the appearance of the target. The automatic spreading of activation is proposed to be measured with a short SOA (from 150 to 500 ms) (ALTARRIBA et BASNIGHT-BROWN, 2007, NEELY, 1977 , NEELY, 1991).

In the presence of a long SOA (over 500 ms), the process is no longer considered purely automatic and falls within the framework of the expectancy-based priming theory (Becker, 1980). Beyond 500 ms, the prime and target are assumed to be fully processed, allowing the reader to consciously establish a connection between them, indicating that a selection process has also been completed (Holderbaum et al., 2019). Therefore, the facilitation effect can still be observed, but it includes both automatic activation and an additional process, which was highlighted by Copland et al. (2007) and will be explored here as an expression of an early selection process (inhibition of internal activation).

Indeed, using fMRI observations, COPLAND et al. (2003) and COPLAND et al. (2007) conducted a primed lexical decision task (LDT) with a long SOA (1000 ms) and ambiguous words as primes. The ambiguous words (e.g., right) were related either to a target corresponding to their most commonly used meaning, considered dominant (e.g., correct), or to a target corresponding to their less commonly used meaning, considered subordinate (e.g., direction). Researchers found reduced activity in the middle temporal cortex for dominant pairs, demonstrating a facilitation effect of priming. They also showed that decreased activity in the left inferior prefrontal regions was related to subordinate pairs compared to non-related pairs (COPLAND et al., 2003, COPLAND et al., 2007). They propose this second process as reflecting an inhibition of unrelated information (internally activated).

This finding can be interpreted as evidence of the influence of an ambiguous word, even when it facilitates a response to a concept related to its subordinate meaning. It results in a different effect compared to prime-target pairs that are completely unrelated semantically. This supports the idea that after the automatic spreading of activation from one concept to another, semantic cognition involves a multifunctional executive contribution (BADRE et WAGNER, 2002, BADRE et al., 2005, BINDER et DESAI, 2011, COPLAND et al., 2003, COPLAND et al., 2007, HOFFMAN et MORCOM, 2018), presumably beginning with inhibition.

However, beyond 1200 ms of SOA, the paradigm no longer allows for the observation of automatic semantic activation, as multiple cycles of activation or selection can be processed successively in a controlled manner (NEELY et al., 2010).

	AD	AS	M	NR	ANOVA : F, p-value
all items (N by version)	30 (8)	30 (8)	62 (16)	124 (32)	
Frq. Book (prime)	23.26 (24.70)	24.28 (24.79)	28.26 (35.6)	33.22 (51.65)	F = 0.66, p = 0.55
Frq. Book (target)	48.38 (40.76)	27.99 (25.82)	36.87 (33.79)	34.29 (34.26)	F = 2.08, p = 0.10
Frq. Film (prime)	18.93 (21.33)	17.66 (21.76)	19.77 (28.9)	20.8 (28.9)	F = 0.13, p = 0.94
Frq. Film (target)	63.22 (109)	31.46 (46.45)	42.62 (63.76)	33.08 (56.2)	F = 1.95, p = 0.12
nb letters (prime)	5.59 (1.52)	5.77 (1.62)	6.71 (1.86)	6.38 (1.69)	F = 4.32, p = 0.009
nb letters (target)	6.22 (1.56)	6.03 (1.85)	6.04 (1.85)	6.55 (1.91)	F = 1.40, p = 0.24
Orth.ngb (prime)	3.84 (4.5)	3.38 (4.13)	2.5 (3.75)	2.64 (3.49)	F = 2.63, p = 0.30
Orth.ngb (target)	3.25 (3.70)	3.29 (4)	3.71 (4.18)	2.8 (4.11)	F = 0.67, p = 0.60
concreteness (prime)	3.79 (1.07)	3.74 (1.16)	3.78 (1.10)	3.86 (1.05)	F = 0.08, p = 0.96
concreteness (target)	3.86 (1.06)	3.67 (1.1)	3.70 (1.06)	3.98 (0.90)	F = 1.45, p = 0.23
Semantic association (LSA)	0.34 (0.19)	0.12 (0.07)	0.27 (0.13)	0.1 (0.11)	M, PD < PS, NR : p < 0.001

TABLE 2 – LDT : Psycholinguistic values and semantic associations by conditions on all items

1.2.4 Concreteness

Concreteness refers to the extent to which words refer to something that can be perceived sensorially (seen, heard, touched, smelled or tasted). A concrete representation (e.g. apple) will be processed more quickly and accurately than an abstract representation (e.g. feeling) (BINDER et DESAI, 2011, VERGALLITO et al., 2020), for example in lexical decision tasks (ADORNI et PROVERBIO, 2012, BINDER et DESAI, 2011, PAPAGNO et al., 2009). The effect of concreteness can be seen in several cognitive tasks involving words or pictures. A study by DEL MASCHIO et al. (2021) showed that the effect of this variable was correlated between tasks in which words were used in context or in isolation. Although it is based on all the sensory perceptions of a concept, concreteness is strongly correlated with imageability (PAIVIO et al., 1968), i.e. the ease with which it is possible to form a mental image associated with a word (FERRAND et al., 2018; MICELI et al., 2021).

The effect of age on the concreteness effect is discussed by some authors considering it is influenced by age (MICELI et al., 2021), or not (PAIVIO, 1991; ROXBURY et al., 2016). When available, the concreteness was reported from the norms (for all ages) provided by the Open Lexicon French database (NEW et al., 2004), and coming from BONIN et al. (2018).

For missing values, one hundred and four French-speaking participants (55 women and 49 men), aged between 55 and 75 with a mean level of 13 years of education (SD = 3.74) took part in the focus group for concreteness. We built the task on the psychological testing platform www.testable.com (Rezlescu et al., 2020). The instruction was reproduced from the normative work of Miceli et al. (2021) : We ask you to indicate the degree to which the words presented in the table below refer to concrete concepts. In other words, we ask you to indicate, by ticking the corresponding case in the table, the degree to which the words presented refer to objects, people, places or things that can be seen, heard, tasted, and/or touched. A score of 5 will indicate that the word refers to a very concrete concept, while a score of 1 will indicate that the concept refers to a very abstract concept MICELI et al., 2021).

The values are a mathematical mean of the answers for every item.

1.2.5 Frequency

Frequency is an essential psycholinguistic variable, predictive of the speed at which a word is processed (SOLOMON et HOWES, 1951) and determined by the number of occurrences of a word in a corpus (e.g., books or films). High-frequency words are processed more accurately

and more quickly than low-frequency words, both in comprehension and production (BAAYEN et al., 2006, BALOTA et al., 2004, YAP et BALOTA, 2009)

This frequency effect is observed in several verbal and semantic tasks, such as pronunciation (BALOTA et CHUMBLEY, 1985, FORSTER et CHAMBERS, 1973), categorization (ANDREWS et HEATHCOTE, 2001), and lexical decision. Regarding the latter task, a systematic review of the literature by FERRAND et al. (2018) showed that frequency was the most predictive variable in visual lexical decision tasks.

However, in primed lexical decision tasks (such as the one described in the methodological section), the effect of frequency should be considered in reverse. Indeed, while high-frequency words predict faster responses in lexical decision-making, the opposite is observed for priming effects. Specifically, low-frequency target words elicit a greater priming effect (BECKER et KILLION, 1977, BOROWSKY et BESNER, 1993, PLAUT, 1995), particularly in participants with fewer years of formal education (YAP et BALOTA, 2009).

Frequency values from books and filmographic databases were collected from the New French lexical database Lexique (NEW et al., 2004).

1.2.6 Orthographic Neighborhood

The orthographic neighborhood refers to the set of words that can be formed by changing only one letter of an original word (e.g. French words *Coeur* / *Soeur*) (COLTHEART et al., 1977). Its size influences the cognitive processing of the word, but there is no complete consensus on the meaning of this effect (HUNTSMAN et LIMA, 2002). Several authors have suggested that the size of the orthographic neighborhood facilitates word identification (LUCE, 1986) and naming tasks (GUNTHER et GREASE, 1985, LAXON et al., 1988, SCHEERER, 1987). The size of the orthographic neighborhood can play a facilitating role in the word recognition process, see HUNTSMAN et LIMA (2002). SNODGRASS et MINTZER (1993), on the other hand, reported an inhibitory effect of the orthographic neighborhood in whole-word or fragmented-form identification tasks. The logic underlying a facilitating effect of the orthographic neighborhood is that of multiple activation. The larger the orthographic neighborhood, the more simultaneous activations of concepts will be created, the accumulation of which will result in a facilitating effect (ANDREWS, 1989, ANDREWS, 1992, ANDREWS, 1997). Following the same logic, this effect may be accompanied by an inhibitory effect of the more frequent orthographic neighbor (French word *maison*; frequency of 461) on a less frequent original word (French word *saison*; frequency of 35) (HUNTSMAN et LIMA, 2002, SNODGRASS et MINTZER, 1993). This means that the existence of a very frequent orthographic neighbor can slow down the processing of the original word. Value of orthographic neighborhood were collected from the New French lexical database Lexique (NEW et al., 2004).

1.2.7 Semantic association

The strength of semantic link for the counterbalances pairs of the task (AD, AS, M, NR) were reported from the latent semantic association value (LSA) database (LANDAUER et al., 1998). LSA is an index calculated by a matrix factorization technique performed on a vast text corpus. The LSA value for a pair of items represents the strength of their estimate semantic relationship, from 0 to 1, 1 being the strongest association possible.

In the experimental tasks of the present research, the values of semantic links are determined on the basis of the values provided by the Latent Semantic Analysis (LSA) model (LANDAUER et al., 1998). LSA is a model Hyperspace Analogue to Language that reflects two concepts consistent with the COLLINS et LOFTUS (1975) model, namely semantic distance and semantic

relation. In the sense of COLLINS et LOFTUS (1975) semantic distance is the length of the shortest path (direct or indirect) between two elements within the semantic network (e.g. "dog" is closer to "cat" than to "table", HUTCHISON, 2003) and semantic relatedness reflects the strength of the connection which is measured according to the number of intermediate connections between two concepts in the network. Consequently, "cat" may be at the same semantic distance from "bird" as from "dog", but the semantic relationship will be stronger between "cat" and "dog" because beyond their direct link, they also share several indirect semantic paths. The HAL model effectively captures both the semantic distance and the semantic relationship, by simultaneously analyzing the frequency of co-occurrences and the Euclidean distance (i.e. the straight line between two vectors) between two words, in a multidimensional space (KUMAR, 2021).

The validity of the HAL model for determining a semantic relationship value between two concepts has been discussed. LIVESAY et BURGESS (1998) indicated that it was unable to account for mediated priming (lion - tiger - stripe), but CHWILLA et KOLK (2002) obtained evidence that initiate-target stimuli, mediated by two to three intermediaries, were semantically more linked according to LSA values than were unmatched control pairs. HUTCHISON (2003) also reproduced this control for mediated semantic values with the LSA. Finally, models based on hyperspace of language accurately predict priming effects for relations that are semantic and associative, but not for solely associative relations (HUTCHISON, 2003).

TABLE 3 – LDT : Items of all versions

VERSION	Condition	prime	target
1	Monosemic related	contribuable	impôt
1	Monosemic related	choppe	bière
1	Monosemic related	poêle	cuisine
1	Monosemic related	conte	fable
1	Monosemic related	accuse	juge
1	Monosemic related	larve	papillon
1	Monosemic related	lettre	poste
1	Monosemic related	filiale	société
1	Monosemic related	couture	aiguille
1	Monosemic related	dents	vampire
1	Monosemic related	lunette	vision
1	Monosemic related	image	reflet
1	Monosemic related	parfum	vanille
1	Monosemic related	religion	seigneur
1	Monosemic related	flaque	boue
1	Monosemic related	tonalite	parole
1	Ambiguous - dominant acceptance	stade	football
1	Ambiguous - dominant acceptance	glace	miroir
1	Ambiguous - dominant acceptance	note	musique
1	Ambiguous - dominant acceptance	pensée	opinion
1	Ambiguous - dominant acceptance	solution	problème
1	Ambiguous - dominant acceptance	page	roman
1	Ambiguous - dominant acceptance	navet	soupe
1	Ambiguous - dominant acceptance	canne	sucré
1	Ambiguous - subordinated acceptance	bidet	âne

Continued on next page

VERSION	Condition	prime	target
1	Ambiguous - subordinated acceptance	fléau	arme
1	Ambiguous - subordinated acceptance	arête	carre
1	Ambiguous - subordinated acceptance	ampoule	cloaque
1	Ambiguous - subordinated acceptance	couette	coiffure
1	Ambiguous - subordinated acceptance	accès	douleur
1	Ambiguous - subordinated acceptance	aube	prêtre
1	Ambiguous - subordinated acceptance	arc	triomphe
1	Counterbalanced non-related	absent	fleur
1	Counterbalanced non-related	baleine	cahier
1	Counterbalanced non-related	ballon	parapluie
1	Counterbalanced non-related	bibliothèque	nation
1	Counterbalanced non-related	canapé	explication
1	Counterbalanced non-related	chat	paupière
1	Counterbalanced non-related	concept	literie
1	Counterbalanced non-related	douceur	nul
1	Counterbalanced non-related	église	pêche
1	Counterbalanced non-related	étudiant	poisson
1	Counterbalanced non-related	évier	médecin
1	Counterbalanced non-related	garrot	entrée
1	Counterbalanced non-related	indice	coussin
1	Counterbalanced non-related	instrument	légume
1	Counterbalanced non-related	mandoline	crépuscule
1	Counterbalanced non-related	ombrelle	divan
1	Counterbalanced non-related	physique	méditation
1	Counterbalanced non-related	pop-corn	index
1	Counterbalanced non-related	pupille	raisonnement
1	Counterbalanced non-related	réflexion	fromage
1	Counterbalanced non-related	repas	lampe
1	Counterbalanced non-related	résolution	plumard
1	Counterbalanced non-related	salade	fauteuil
1	Counterbalanced non-related	salon	os
1	Counterbalanced non-related	soulage	flèche
1	Counterbalanced non-related	succession	peste
1	Counterbalanced non-related	trousse	pétale
1	Counterbalanced non-related	tulipe	souris
1	Counterbalanced non-related	union	vue
1	Counterbalanced non-related	vase	cabinet
2	Monosemic related	divan	fauteuil
2	Monosemic related	lecture	roman
2	Monosemic related	cabinet	médecin
2	Monosemic related	mer	poisson
2	Monosemic related	méditation	raisonnement
2	Monosemic related	fromage	souris
2	Monosemic related	paupière	vue
2	Monosemic related	averse	parapluie
2	Monosemic related	question	explication
2	Monosemic related	pays	nation

Continued on next page

VERSION	Condition	prime	target
2	Monosemic related	laine	aiguille
2	Monosemic related	pop-corn	film
2	Monosemic related	salade	légume
2	Monosemic related	tulipe	pétale
2	Monosemic related	zéro	nul
2	Ambiguous - dominant acceptance	couette	coussin
2	Ambiguous - dominant acceptance	aube	crépuscule
2	Ambiguous - dominant acceptance	accès	entrée
2	Ambiguous - dominant acceptance	arc	flèche
2	Ambiguous - dominant acceptance	ampoule	lampe
2	Ambiguous - dominant acceptance	arête	os
2	Ambiguous - dominant acceptance	fléau	peste
2	Ambiguous - dominant acceptance	bidet	toilette
2	Ambiguous - dominant acceptance	perception	vision
2	Ambiguous - dominant acceptance	recette	argent
2	Ambiguous - subordinated acceptance	solution	chimie
2	Ambiguous - subordinated acceptance	stade	étape
2	Ambiguous - subordinated acceptance	note	examen
2	Ambiguous - subordinated acceptance	pensée	fleur
2	Ambiguous - subordinated acceptance	canne	pêche
2	Ambiguous - subordinated acceptance	page	seigneur
2	Counterbalanced non-related	glace	accuse
2	Counterbalanced non-related	siège	chope
2	Counterbalanced non-related	chenille	conte
2	Counterbalanced non-related	légende	contribuable
2	Counterbalanced non-related	navet	parole
2	Counterbalanced non-related	avocat	conte
2	Counterbalanced non-related	bide	lettre
2	Counterbalanced non-related	timbre	poêle
2	Counterbalanced non-related	église	impôt
2	Counterbalanced non-related	étudiant	bière
2	Counterbalanced non-related	physique	miroir
2	Counterbalanced non-related	salon	cloque
2	Counterbalanced non-related	selle	coiffure
2	Counterbalanced non-related	soulage	poste
2	Counterbalanced non-related	succession	bière
2	Counterbalanced non-related	tresse	cuisine
2	Counterbalanced non-related	matelas	prêtre
2	Counterbalanced non-related	lumière	puceron
2	Counterbalanced non-related	sortie	prêtre
2	Counterbalanced non-related	carquois	coiffure
2	Counterbalanced non-related	pied	filiale
2	Counterbalanced non-related	rat	planète
2	Counterbalanced non-related	tube	quartier
2	Counterbalanced non-related	flaque	football
2	Counterbalanced non-related	meurtre	miroir
2	Counterbalanced non-related	moteur	musique

Continued on next page

VERSION	Condition	prime	target
2	Counterbalanced non-related	mulet	opinion
2	Counterbalanced non-related	parade	sucré
2	Counterbalanced non-related	pièce	terme
2	Counterbalanced non-related	rectangle	douleur
3	Monosemic related	matelas	coussin
3	Monosemic related	lumière	crépuscule
3	Monosemic related	sortie	entrée
3	Monosemic related	carquois	flèche
3	Monosemic related	pied	lampe
3	Monosemic related	chien	os
3	Monosemic related	rat	peste
3	Monosemic related	bain	toilette
3	Monosemic related	église	prêtre
3	Monosemic related	étudiant	examen
3	Monosemic related	orteil	cloaque
3	Monosemic related	salon	coiffure
3	Monosemic related	selle	cheval
3	Monosemic related	soulagement	douleur
3	Monosemic related	succession	étape
3	Monosemic related	mélange	chimie
3	Ambiguous - dominant acceptance	canapé	fauteuil
3	Ambiguous - dominant acceptance	vase	fleur
3	Ambiguous - dominant acceptance	garrot	médecin
3	Ambiguous - dominant acceptance	page	roman
3	Ambiguous - dominant acceptance	baleine	poisson
3	Ambiguous - dominant acceptance	stade	football
3	Ambiguous - dominant acceptance	chat	souris
3	Ambiguous - dominant acceptance	pupille	vue
3	Ambiguous - subordinated acceptance	glace	vanille
3	Ambiguous - subordinated acceptance	siège	auto
3	Ambiguous - subordinated acceptance	chenille	char
3	Ambiguous - subordinated acceptance	légende	explication
3	Ambiguous - subordinated acceptance	navet	film
3	Ambiguous - subordinated acceptance	avocat	légume
3	Ambiguous - subordinated acceptance	bide	nul
3	Ambiguous - subordinated acceptance	timbre	parole
3	Counterbalanced non-related	bidet	concept
3	Counterbalanced non-related	fléau	ballon
3	Counterbalanced non-related	arête	bibliothèque
3	Counterbalanced non-related	ampoule	douceur
3	Counterbalanced non-related	couette	adhésion
3	Counterbalanced non-related	accès	évier
3	Counterbalanced non-related	aube	instrument
3	Counterbalanced non-related	arc	résolution
3	Counterbalanced non-related	flaque	football
3	Counterbalanced non-related	fromage	roman
3	Counterbalanced non-related	meurtre	miroir

Continued on next page

VERSION	Condition	prime	target
3	Counterbalanced non-related	moteur	feuille
3	Counterbalanced non-related	mulet	opinion
3	Counterbalanced non-related	parade	miroir
3	Counterbalanced non-related	pièce	opinion
3	Counterbalanced non-related	rectangle	musique
3	Counterbalanced non-related	cabinet	boue
3	Counterbalanced non-related	divan	apéritif
3	Counterbalanced non-related	usage	boue
3	Counterbalanced non-related	litterie	arme
3	Counterbalanced non-related	méditation	âne
3	Counterbalanced non-related	mer	groseille
3	Counterbalanced non-related	paupière	marbre
3	Counterbalanced non-related	pétale	mission
3	Counterbalanced non-related	couture	argent
3	Counterbalanced non-related	dents	carotte
3	Counterbalanced non-related	lunette	bière
3	Counterbalanced non-related	chemin	mission
3	Counterbalanced non-related	parfum	genou
3	Counterbalanced non-related	religion	acier
3	Counterbalanced non-related	succès	copain
3	Counterbalanced non-related	tonalite	société
4	Monosemic related	ballon	football
4	Monosemic related	bibliothèque	roman
4	Monosemic related	évier	miroir
4	Monosemic related	instrument	musique
4	Monosemic related	concept	opinion
4	Monosemic related	résolution	problème
4	Monosemic related	repas	soupe
4	Monosemic related	douceur	sucré
4	Monosemic related	il	vue
4	Monosemic related	fromage	souris
4	Monosemic related	meurtre	arme
4	Monosemic related	moteur	auto
4	Monosemic related	mulet	âne
4	Monosemic related	parade	char
4	Monosemic related	pièce	argent
4	Monosemic related	rectangle	carre
4	Ambiguous - dominant acceptance	bide	bière
4	Ambiguous - dominant acceptance	recette	cuisine
4	Ambiguous - dominant acceptance	légende	fable
4	Ambiguous - dominant acceptance	avocat	juge
4	Ambiguous - dominant acceptance	chenille	papillon
4	Ambiguous - dominant acceptance	timbre	poste
4	Ambiguous - dominant acceptance	siège	société
4	Ambiguous - subordinated acceptance	perception	impôt
4	Ambiguous - subordinated acceptance	chas	aiguille
4	Ambiguous - subordinated acceptance	canapé	apéritif

Continued on next page

VERSION	Condition	prime	target
4	Ambiguous - subordinated acceptance	vase	boue
4	Ambiguous - subordinated acceptance	garrot	cheval
4	Ambiguous - subordinated acceptance	pupille	nation
4	Ambiguous - subordinated acceptance	baleine	parapluie
4	Ambiguous - subordinated acceptance	réflexion	reflet
4	Ambiguous - subordinated acceptance	couette	coiffure
4	Counterbalanced non-related	solution	avenue
4	Counterbalanced non-related	stade	bain
4	Counterbalanced non-related	note	carquois
4	Counterbalanced non-related	pensée	marbre
4	Counterbalanced non-related	canne	matelas
4	Counterbalanced non-related	page	alcool
4	Counterbalanced non-related	couette	rat
4	Counterbalanced non-related	bidet	sortie
4	Counterbalanced non-related	couture	coussin
4	Counterbalanced non-related	dents	crépuscule
4	Counterbalanced non-related	lunette	entrée
4	Counterbalanced non-related	chaîne	poivre
4	Counterbalanced non-related	parfum	crépuscule
4	Counterbalanced non-related	religion	entrée
4	Counterbalanced non-related	succès	flèche
4	Counterbalanced non-related	tonalité	os
4	Counterbalanced non-related	accuse	planète
4	Counterbalanced non-related	chope	poivre
4	Counterbalanced non-related	conte	vision
4	Counterbalanced non-related	contribuable	panthère
4	Counterbalanced non-related	filiale	vanille
4	Counterbalanced non-related	larve	seigneur
4	Counterbalanced non-related	lettre	triomphe
4	Counterbalanced non-related	poêle	médecin
4	Counterbalanced non-related	absent	opéra
4	Counterbalanced non-related	indice	médecin
4	Counterbalanced non-related	mandoline	canard
4	Counterbalanced non-related	ombrelle	écureuil
4	Counterbalanced non-related	pop-corn	émeraude
4	Counterbalanced non-related	salade	zinc
4	Counterbalanced non-related	tulipe	acier
4	Counterbalanced non-related	union	raisonnement
	Non-word	four	pitkm
	Non-word	culotte	afellpe
	Non-word	métro	fetrute
	Non-word	branche	sijhge
	Non-word	jupe	factare
	Non-word	analyse	ruthme
	Non-word	trèfle	mouctache
	Non-word	café	pyrimide
	Non-word	sable	holte

Continued on next page

VERSION	Condition	prime	target
	Non-word	souche	clivoge
	Non-word	sympathie	divarsite
	Non-word	température	lutige
	Non-word	corne	verute
	Non-word	focale	pulitesse
	Non-word	radiateur	ginou
	Non-word	pavillon	brucelet
	Non-word	statue	faravent
	Non-word	garantie	chalpignon
	Non-word	récif	milange
	Non-word	digestif	tricharie
	Non-word	miel	vinuit
	Non-word	séance	chemunee
	Non-word	signe	particune
	Non-word	zombie	saluire
	Non-word	officiel	isalant
	Non-word	livreur	cabarat
	Non-word	écaille	paxience
	Non-word	publicité	dintier
	Non-word	trimestre	noctirne
	Non-word	croyance	tallette
	Non-word	figurine	genepal
	Non-word	bouton	gratitute
	Non-word	message	tamboir
	Non-word	malice	dosour
	Non-word	réveillon	imprumante
	Non-word	opticien	getiteur
	Non-word	empire	accobate
	Non-word	bronzage	coqualle
	Non-word	universel	gelagine
	Non-word	brasserie	clevasse
	Non-word	herbe	citoyan
	Non-word	primate	cauleur
	Non-word	chaussette	carnoge
	Non-word	crépi	roudine
	Non-word	lamelle	souroupe
	Non-word	risque	fantoisie
	Non-word	évasion	methide
	Non-word	énigme	sagense
	Non-word	charlatan	confurt
	Non-word	voyou	sevatif
	Non-word	économie	peraple
	Non-word	moutarde	securide
	Non-word	croquette	transpert
	Non-word	épuiette	coramel
	Non-word	destin	epigard
	Non-word	réaction	piauvre

Continued on next page

VERSION	Condition	prime	target
	Non-word	matière	ritraite
	Non-word	biscuit	speviale
	Non-word	limonade	sarapace
	Non-word	détente	recalte
	Non-word	cachette	resonarce
	Non-word	lessive	naid
	Non-word	pompe	marsure
	Non-word	choix	prifil
	Non-word	entracte	evincelle
	Non-word	reptile	colve
	Non-word	phoque	bauchier
	Non-word	évidence	caprite
	Non-word	clou	budgit
	Non-word	télévision	epentail
	Non-word	courage	repege
	Non-word	hurlement	mesu
	Non-word	aide	mipeur
	Non-word	vitesse	pudour
	Non-word	cape	optimisse
	Non-word	formule	sigrature
	Non-word	noyau	drainave
	Non-word	propriété	soumon
	Non-word	acteur	sylfege
	Non-word	valse	setaine
	Non-word	ricochet	ficotine
	Non-word	vestige	monsueur
	Non-word	vermine	saupe
	Non-word	endroit	echinge
	Non counterbalanced fill-in pair	pieu	valise
	Non counterbalanced fill-in pair	selle	film
	Non counterbalanced fill-in pair	recette	argent
	Non counterbalanced fill-in pair	literie	plumard
	Non counterbalanced fill-in pair	chien	douleur
	Non counterbalanced fill-in pair	visage	musique
	Non counterbalanced fill-in pair	orteil	cloaque
	Non counterbalanced fill-in pair	terrier	élève
	Non counterbalanced fill-in pair	ruche	bal
	Non counterbalanced fill-in pair	gland	flamme
	Non counterbalanced fill-in pair	trompe	douche
	Non counterbalanced fill-in pair	pince	mine
	Non counterbalanced fill-in pair	vigne	jambe
	Non counterbalanced fill-in pair	étang	arbre
	Non counterbalanced fill-in pair	robinet	salon
	Non counterbalanced fill-in pair	ski	trou
	Non counterbalanced fill-in pair	pain	papier
	Non counterbalanced fill-in pair	singe	écran
	Non counterbalanced fill-in pair	piège	fenêtre

Continued on next page

VERSION	Condition	prime	target
	Non counterbalanced fill-in pair	dinette	arbalète
	Non counterbalanced fill-in pair	spatule	hautbois
	Non counterbalanced fill-in pair	ventilateur	cartouche
	Non counterbalanced fill-in pair	short	capot
	Non counterbalanced fill-in pair	scie	tige
	Non counterbalanced fill-in pair	flute	rond
	Non counterbalanced fill-in pair	casserole	sandale
	Non counterbalanced fill-in pair	crayon	nez
	Non counterbalanced fill-in pair	terrier	élève
	Non counterbalanced fill-in pair	ruche	bal
	Non counterbalanced fill-in pair	gland	flamme
	Non counterbalanced fill-in pair	trompe	douche
	Non counterbalanced fill-in pair	pince	mine
	Non counterbalanced fill-in pair	vigne	jambe
	Non counterbalanced fill-in pair	étang	arbre
	Non counterbalanced fill-in pair	robinet	salon
	Non counterbalanced fill-in pair	ski	trou
	Non counterbalanced fill-in pair	pain	papier
	Non counterbalanced fill-in pair	singe	écran
	Non counterbalanced fill-in pair	piège	fenêtre
	Non counterbalanced fill-in pair	dinette	arbalète
	Non counterbalanced fill-in pair	spatule	hautbois
	Non counterbalanced fill-in pair	ventilateur	cartouche
	Non counterbalanced fill-in pair	short	capot
	Non counterbalanced fill-in pair	scie	tige
	Non counterbalanced fill-in pair	flute	rond
	Non counterbalanced fill-in pair	casserole	sandale
	Non counterbalanced fill-in pair	crayon	nez
	Non counterbalanced fill-in pair	terrier	élève
	Non counterbalanced fill-in pair	ruche	bal
	Non counterbalanced fill-in pair	gland	flamme
	Non counterbalanced fill-in pair	trompe	douche
	Non counterbalanced fill-in pair	pince	mine
	Non counterbalanced fill-in pair	vigne	jambe
	Non counterbalanced fill-in pair	étang	arbre
	Non counterbalanced fill-in pair	robinet	salon
	Non counterbalanced fill-in pair	ski	trou
	Non counterbalanced fill-in pair	pain	papier
	Non counterbalanced fill-in pair	singe	écran
	Non counterbalanced fill-in pair	piège	fenêtre
	Non counterbalanced fill-in pair	dinette	arbalète
	Non counterbalanced fill-in pair	spatule	hautbois
	Non counterbalanced fill-in pair	ventilateur	cartouche
	Non counterbalanced fill-in pair	short	capot
	Non counterbalanced fill-in pair	scie	tige
	Non counterbalanced fill-in pair	flute	rond
	Non counterbalanced fill-in pair	casserole	sandale

Continued on next page

VERSION	Condition	prime	target
	Non counterbalanced fill-in pair	crayon	nez
	Non counterbalanced fill-in pair	terrier	élève
	Non counterbalanced fill-in pair	ruche	bal
	Non counterbalanced fill-in pair	gland	flamme
	Non counterbalanced fill-in pair	trompe	douche
	Non counterbalanced fill-in pair	pince	mine
	Non counterbalanced fill-in pair	vigne	jambe
	Non counterbalanced fill-in pair	étang	arbre
	Non counterbalanced fill-in pair	robinet	salon
	Non counterbalanced fill-in pair	ski	trou
	Non counterbalanced fill-in pair	pain	papier
	Non counterbalanced fill-in pair	singe	écran
	Non counterbalanced fill-in pair	piège	fenêtre
	Non counterbalanced fill-in pair	dinette	arbalète
	Non counterbalanced fill-in pair	spatule	hautbois
	Non counterbalanced fill-in pair	ventilateur	cartouche
	Non counterbalanced fill-in pair	short	capot
	Non counterbalanced fill-in pair	scie	tige
	Non counterbalanced fill-in pair	flute	rond
	Non counterbalanced fill-in pair	casserole	sandale

2 CTTAT

This experimental task is based on requesting participants to pair, for each trial, two words. In each trial, they are presented with one cue word and must select from several options (2 or 4) which word satisfies the association instruction. The association instruction varies; it could involve selecting the word with the same general meaning (e.g., seat - chair), or the word representing something of the same color (e.g., strawberry, ladybug), or of the same size (e.g., tooth, pebble).

2.1 Design

Our task is freely inspired by existing tasks aimed to enhance the selection and the controlled semantic retrieval (BADRE et al., 2005, BADRE et WAGNER, 2002, HOFFMAN et MORCOM, 2018, MOSS et al., 2005, THOMPSON-SCHILL et al., 2005, THOMPSON-SCHILL et al., 1997, THOMPSON-SCHILL et al., 1998). Based on these existing task, our differences are two.

First, we replaced the condition of strong/weak semantic associate by a condition of association of a ambiguous word with a word corresponding to its dominant or subordinated meaning.

Secondly, we balanced the presence/absence of a distractor semantically related to the cue for every condition of association. All the previous task were in English, we composed our original corpus of French words.

The cue to target association task (CTTAT) was designed in two versions different only on the number of distractors (choosing the answer among 2 or 4 words). The task was made with words and composed of 3 blocks in which the instruction was, based on a cue, to choose a target (1) corresponding the most to an equal meaning of the cue (condition synonym), or (2) having the same size or (3) color.

For the Synonym condition, participants were instructed to choose the target word representing most closely the same meaning of the cue word. To distinguish the target from semantically related distractors, the instruction voiced that the chosen word does not have to be only related to the cue (as caterpillar and butterfly) but to be the word that is the closest to representing the same meaning (as caterpillar and centipede). Twenty-eight ambiguous words were token in the corpus of the Lexical decision task to create this condition of similarity of global meaning. Fourteen were linked to a target equivalent to their dominant meaning and fourteen were linked to a target equivalent to their subordinated meaning.

CONDITION OF ASSOCIATION 1 and 2

You will see a main word displayed at the top of the screen, followed by four other words below it. Your task is to select the word whose meaning is closest to that of the main word. The two words do not always have to be perfect synonyms, but they should make sense in a sentence such as 'WORD1 is a bit like WORD2' or 'WORD1 has almost the same meaning as WORD2'. Your choice should reflect the word with the closest meaning, not just one that is generally related to the main word. There is always one correct answer. Let's begin with a few practice trials."

(dominant) PERCEPTION (sensation/ opinion)	
(1) FONCTIONNAIRE (civil servant)	(2) ROSIER (rosebush)
(3) POUPÉE (doll)	(4) SENSATION (sensation)
The answer is "sensation" (sensation) The distractor "fonctionnaire" (civil servant) is semantically related to the clue.	

(subordinate) GLACE (mirror / ice cream)	
(1) SORBET (sorbet)	(2) CIGARE (cigar)
(3) PORTRAIT (portrait)	(4) HOUX (holly)
The answer is "sorbet" (sorbet). The distractors are not related to the clue	

CONDITION OF ASSOCIATION 3 and 4

Association feature-based

Instruction for FIND THE SAME SIZE: "You will see a main word displayed at the top of the screen, followed by four other words below it. Your task is to choose the word that refers to something closest in size to the main word. There is always one correct answer."

Instruction for FIND THE SAME COLOR: "You will see a main word displayed at the top of the screen, followed by four other words below it. Your task is to choose the word that refers to something with the same color as the main word. There is always one correct answer."

"Let's begin with a few practice trials."

(size) CAILLOU (pebble)	
(1) FILLETTE (little girl)	(2) PHONOGRAPHE (phonograph)
(3) INFORMATION (information)	(4) DENT (tooth)
The correct answer is "dent" (tooth) because it has the same size of a "caillou" (pebble). The distractors are not related to the cue	

(color) COCCINELLE (ladybug)	
(1) PUCERON (aphid)	(2) STEREO (stereo)
(3) COMPLICITÉ (complicity)	(4) GROSEILLE (currant)
The correct answer is "groseille" (currant). The distractor "puceron" (aphid) is semantically related to the cue.	

FIGURE 2 – cue to target association task : examples of trials

For each condition of association, in half of the trials, the foil was semantically associated with the cue (semantic link condition) but unfit to correspond to the instruction, enhancing the selection demand. For example, in the proposition [cue] caterpillar [target] centipede [distractor] butterfly, centipede is almost a synonym of caterpillar, while butterfly is semantically linked to caterpillar but does not have the same meaning. In the other half, the foil was incongruent. Table 4 illustrate the despatching of trials among conditions.

All words are presented in table 6, while, in this document, table 5 presents the mean and standard deviations of concreteness, imageability, frequency, length, and LSA strength of association for all categories of words in the tasks (cue, target, foil, and for the 4-item version, word 3 and word 4).

Considering the combination of conditions of meaning and the number of words to associate, it was almost impossible to homogeneously balance these values between the conditions. Therefore, they are measured as covariates in the statistical analysis of the results.

Association $\rightarrow$	Syn. dom.	Syn. sub.	size	color
All non-related	7 trials	7 trials	8 trials	8 trials
One semantically related to cue	7 trials	7 trials	8 trials	8 trials

TABLE 4 – Trial distribution based on association type and distractor condition.

TABLE 5 – cue to target association task : psycholinguistic values

	cue	Target	Foil	Word 3	Word 4
Concreteness	4.20 (0.84)	3.68 (0.83)	4.18 (0.71)	3.66 (1.02)	3.35 (0.98)
Frequency	18.53 (22.95)	18.82 (28.32)	17.26 (16.12)	18.93 (3.1)	4.97 (6.8)
Imageability	4.58 (0.65)	4.22 (0.96)	4.09 (0.89)	4.06 (0.83)	3.48 (1.03)
Length	6.36 (1.73)	7.68 (8.5)	7.18 (2.1)	6.21 (1.55)	7 (1.76)
LSA cue-foil	0.16 (0.19)				
LSA cue-target	0.13 (0.12)				

cue-target	cue	EN. cue	Target	EN. target	cue-foil	Foil	EN. Foil	Word 3	Word 4
amb-dom	ampoule	BULB	lampe	LAMP	semantic link	clarté	CLARITY	pompe	torero
amb-dom	arc	BOW	voûte	ARCH	semantic link	plafond	CEILING	crédit	lynx
amb-dom	baleine	WHALE	tige	ROD	semantic link	soutien-gorge	BRA	wagon	praticien
amb-dom	bidet	BIDET	cuvette	BOWL	semantic link	blanc	WHITE	congé	figue
amb-dom	canne	CANE	bâton	STICK	semantic link	vieillesse	ELDERLY	tonnerre	poulpe
amb-dom	couette	PONY TAIL	coiffure	HAIRSTYLE	semantic link	brosse	BRUSH	trouble	prunier
amb-dom	garrot	TOURNIQUET	encolure	NECKLINE	semantic link	équitation	RIDING	nappe	loutre
amb-dom	boulon	BOLT	fable	FABLE	Incongruent	photographie	PHOTOGRAPHY	ciment	boulon
amb-dom	page	PAGE	feuille	LEAF	Incongruent	château	CASTLE	branche	laitier
amb-dom	perception	PERCEPTION	sensation	SENSATION	Incongruent	fonctionnaire	OFFICIAL	poupée	rosier
amb-dom	pupille	PUPIL	orphelin	ORPHAN	Incongruent	lunette	LUNETTE	issue	stèle
amb-dom	réflexion	REFLECTION	image	IMAGE	Incongruent	souhait	WISH	mèche	pétale
amb-dom	solution	SOLUTION	dénouement	OUTCOME	Incongruent	magicien	MAGICIAN	torse	orfèvre
amb-dom	timbre	STAMP	sonorité	SOUND	Incongruent	envoi	DELIVERY	royaume	butoir
amb-dom	accès	ACCESS	entrée	INPUT	semantic link	poignée	HANDLEBAR	dentelle	savon
amb-sub	arête	EDGE	ligne	LINE	semantic link	poisson	FISH	baron	haricot
amb-sub	aube	VANE	robe	DRESS	semantic link	prêtre	PRIEST	mousson	bidule
amb-sub	bide	BELLY	échec	FAILURE	semantic link	examen	EXAMINATION	roue	béotien
amb-sub	canapé	SOFA	apéritif	APERITIF	semantic link	coupe	CUT	station	tapissier
amb-sub	chenille	CATERPILLAR	mille-pattes	CENTIPEDE	semantic link	papillon	BUTTERFLY	exercice	tonnelet
amb-sub	fléau	FLAIL	calamité	CALAMITY	semantic link	climat	CLIMATE	rampe	gélatine
amb-sub	glace	ICE	sorbet	SORBET	Incongruent	portrait	PORTRAIT	cigare	houx
amb-sub	note	NOTE	croche	CROCHE	Incongruent	test	TEST	gardien	cagoule
amb-sub	pieu	STAKE	pique	PICK	Incongruent	coussin	CUSHION	secteur	détroit
amb-sub	recette	RECIPE	méthode	METHOD	Incongruent	dette	DEBT	aiguille	luge
amb-sub	siège	SEAT	chaise	CHAIR	Incongruent	contrat	CONTRACT	cirque	orchidée
amb-sub	stade	STADIUM	étape	STEP	Incongruent	football	FOOTBALL	mouche	caracole
amb-sub	vase	VASE	gadoue	SLUSH	Incongruent	plante	PLANT	parquet	diapason
Color	chocolat	CHOCOLATE	ours	BEAR	semantic link	confiserie	CONFISERY	jeudi	zéro
Color	étoile	STAR	citron	LEMON	semantic link	planète	PLANET	urgence	briquette
Color	poireau	LEEK	émeraude	EMERALD	semantic link	tomate	TOMATO	commun	rapine
Color	poivre	PEPPER	pigeon	PIGEON	semantic link	sel	SALT	simplicité	favori
Color	renard	FOX	potiron	PUMPKIN	semantic link	malin	MALIN	fonction	snob
Color	coccinelle	LADYBUG	groseille	CURRENT	semantic link	puceron	PUCERON	complicité	stéréo
Color	cygne	SWAN	lavabo	SINK	semantic link	étang	POND	collection	braderie
Color	menthe	MINT	lézard	LIZARD	semantic link	lavande	LAVENDER	estime	troupeau
Color	fraise	STRAWBERRY	coquelicot	POPPY	Incongruent	écran	SCREEN	vapeur	officiel
Color	feuillage	FOLIAGE	crapaud	TOAD	Incongruent	agrafeuse	SCRATCHER	poésie	boutade
Color	neige	SNOW	crème	CREAM	Incongruent	bouc	CANDLE	pré	saisie
Color	canari	CANARY	banane	BANANA	Incongruent	ruche	RUCHE	piège	rangement
Color	myrtille	BLUEBERRY	encre	INK	Incongruent	radiateur	RADIATOR	demeure	bedaine
Color	sapin	FIR TREE	tortue	TURTLE	Incongruent	mètre	METER	grenier	biais
Color	mandarine	MANDARIN	tigre	TIGER	Incongruent	prise	SOCKET	gamin	natation
Color	réglisse	LIQUORICE	panthère	PANTHER	Incongruent	singe	MONKEY	sergent	collecte
Size	clocher	BELL TOWER	automobile	CAR	semantic link	cantique	CANTIQUE	oeuf	élément
Size	fourchette	FORK	éventail	FAN	semantic link	cuisinière	KITCHEN	tronc	analgésique
Size	chèvre	GOAT	tabouret	STOOL	semantic link	fromage	CHEESE	crayon	détour
Size	marteau	HAMMER	pavé	PAVING STONE	semantic link	atelier	WORKSHOP	fourrure	contenu
Size	soulier	SHOE	croissant	CROISSANT	semantic link	cordonnier	CORDWORKER	charbon	conséquence
Size	matelas	MATTRESS	kayak	KAYAK	semantic link	bougie	CANDLE	couvert	caporal
Size	moustache	MOUSTACHE	serrure	LOCK	semantic link	chapeau	HAT	lion	blague
Size	villa	VILLA	péniche	BARGE	semantic link	statue	STATUE	rat	abduction
Size	cochon	PIG	baril	BARREL	Incongruent	hêtre	BEECH	jaune	aborigène
Size	enveloppe	ENVELOPE	portefeuille	WALLET	Incongruent	remorque	TRAILER	cerf	aspirateur
Size	noix	WALNUT	chignon	CHIGNON	Incongruent	tracteur	TRACTOR	vocation	refuge
Size	hache	AXE	guitare	GUITAR	Incongruent	aquarium	AQUARIUM	bible	plastique
Size	fiole	FLASK	souris	MOUSE	Incongruent	supermarché	SUPERMARKET	panneau	accro
Size	lanterne	LANTERN	magazine	MAGAZINE	Incongruent	chameau	CHAMEAU	airbus	lisière
Size	caillou	PEBBLE	dent	TOOTH	Incongruent	phonographe	PHONOGRAPH	fillette	information
Size	cendrier	ASHTRAY	béret	BERET	Incongruent	parlement	PARLIAMENT	guide	humour

TABLE 6 – CTTAT all items

Statistical analysis

3 Sample Size Justification

3.1 LDT

3.1.1 code R

```
# Define the function for power analysis
power_analysis <- function(n_participants, n_simulations = 20) {
  power_results <- numeric(n_simulations)

  # Estimated effect size
  estimated_f2 <- 0.15 # A moderate effect size

  for (i in 1:n_simulations) {
    # Create a simulated dataset
    set.seed(i) # Ensuring reproducibility
    conditions <- rep(c("Condition1", "Condition2", "
      Condition3", "Condition4"), each = 8)
    subjects <- rep(paste("Subject", 1:n_participants,
      sep = ""), times = 32)
    response_time <- runif(n_participants * length(
      conditions), min = 300, max = 700)

    df <- data.frame(Subject = subjects, Condition =
      conditions, ResponseTime = response_time)

    # Fit the GLMM model
    model <- lmer(ResponseTime ~ Condition + (1 |
      Subject), data = df)

    # Check if the model produces significant
    # coefficients
    fixed_effects <- summary(model)$coefficients
    if (all(is.na(fixed_effects[-1, "Estimate"]))) {
      cat(sprintf("Simulation %d: No significant
        estimates, using default f2 estimate.\n",
          i))
      f2 <- estimated_f2
    } else {
      # Compute effect size f2 (based on model
      # coefficients)
      f2 <- sum(fixed_effects[-1, "Estimate"]^2) /
        (1 - sum(fixed_effects[-1, "Estimate"
          ]^2))
      if (f2 < 0) f2 <- estimated_f2 # Use
        default effect size if f2 is negative
    }

    # Compute power using the effect size
    power_result <- pwr.f2.test(u = 3, v = n_
      participants - 4, f2 = f2, sig.level = 0.05)
    power_results[i] <- power_result$power
  }

  mean_power <- mean(power_results)
  return(mean_power)
```

```

}

# Run power analysis for different sample sizes
participants_list <- c(25, 50, 75, 100)
results <- data.frame(Participants = participants_list, Mean_Power =
  numeric(length(participants_list)))

for (n in participants_list) {
  avg_power <- power_analysis(n)
  results[results$Participants == n, "Mean_Power"] <- avg_
    power
  cat(sprintf("Mean_power_for_%d_participants: %.3f\n", n, avg_
    _power))
}

# Display final results
print(results)

```

3.1.2 Result

```

> print(results)
Participants Mean_Power
1           25  0.2852994
2           50  0.5821296
3           75  0.7895092
4          100  0.9049629

```

3.2 CTTAT

3.2.1 code R

```

install.packages("pwr")
library(pwr)

# Définir les paramètres
effect_size <- 0.25 # Taille de l effet moyen
k <- 2 # Nombre de groupes
m <- 8 # Nombre de mesures répétées
r <- 0.5 # Corrélation entre mesures répétées (estimation)
alpha <- 0.05 # Niveau de signification
power <- 0.80 # Puissance

# Calculer la taille d échantillon
sample_size <- pwr.anova.test(k = k, f = effect_size, sig.level =
  alpha, power = power)
print(sample_size)

```

3.2.2 Result

```

Balanced one-way analysis of variance power calculation
k = 2
n = 63.76561
f = 0.25
sig.level = 0.05
power = 0.8
NOTE: n is number in each group

```

4 LLMs

4.1 LDT - Z scores of RT - Model Diagnostic - LLM - ANOVA - Post-Hoc

The current section was realized following the recommendation of METEYARD et DAVIES, 2020*s best-practice guidance for reporting LLMs.

4.1.1 Assumptions of LMM

Exclusion/ cleaning and transformation of data :

The data of eleven additional participants (ten older, one younger) were excluded due to atypical task behaviors ;

For both tasks, RT is measured only for correct answers and the remaining RT was then screened for outliers using the median absolute deviation (MAD) method (LEYS et al., 2013) placing the outlier threshold at a value of 3, a very conservative threshold under Millers criteria(MILLER, 1991) ;

RT were transformed into z scores (each participant compared to himself) as recommended in LUCE (1986) ;

Model diagnostic :

As presented in figure 3 and figure 4 when conducting an analysis with an F1 approach, where the individual is the statistical unit, we observe that the distribution of response times by condition is influenced by the variable individual. Similarly, when using an F2 approach, where the primetarget pair is the statistical unit, we see that the distribution of response times by condition is influenced by the effect of the primetarget variable. For these reasons, we will test mixed models including condition and group membership as fixed effects, and individuals and primetarget pairs as random effects.

$$m < -lmer(z_{score} \ condition + GROUP + condition * GROUP + (1 + condition|participant) + (1 + condition|prime_{target}), data = data, REML = TRUE)$$

In the initial model specification, we included random intercepts for both participants and primetarget word pairs. However, the rePCA() diagnostics revealed that the variance component for participants was essentially zero (SD 3.0e16), indicating that the random intercept at the participant level did not capture any meaningful variability. This outcome is expected given that the dependent variable was standardized within participants (z-scoring), which removes between-subject differences by construction.

As a consequence, the inclusion of a participant random intercept led to a singular fit, reflecting an overparameterized structure with redundant variancecovariance parameters. According to best practices (METEYARD et DAVIES, 2020), singular fits suggest that the random-effects specification is too complex for the dataset and should be simplified.

Therefore, **we reduced the random-effects structure by retaining only the random intercept for primetarget pairs**, which accounted for a meaningful source of variance (SD 0.36), while excluding the participant-level intercept that contributed no variance. This approach yields a parsimonious model consistent with the data structure, avoids overfitting, and provides stable and interpretable estimates for the fixed effects.

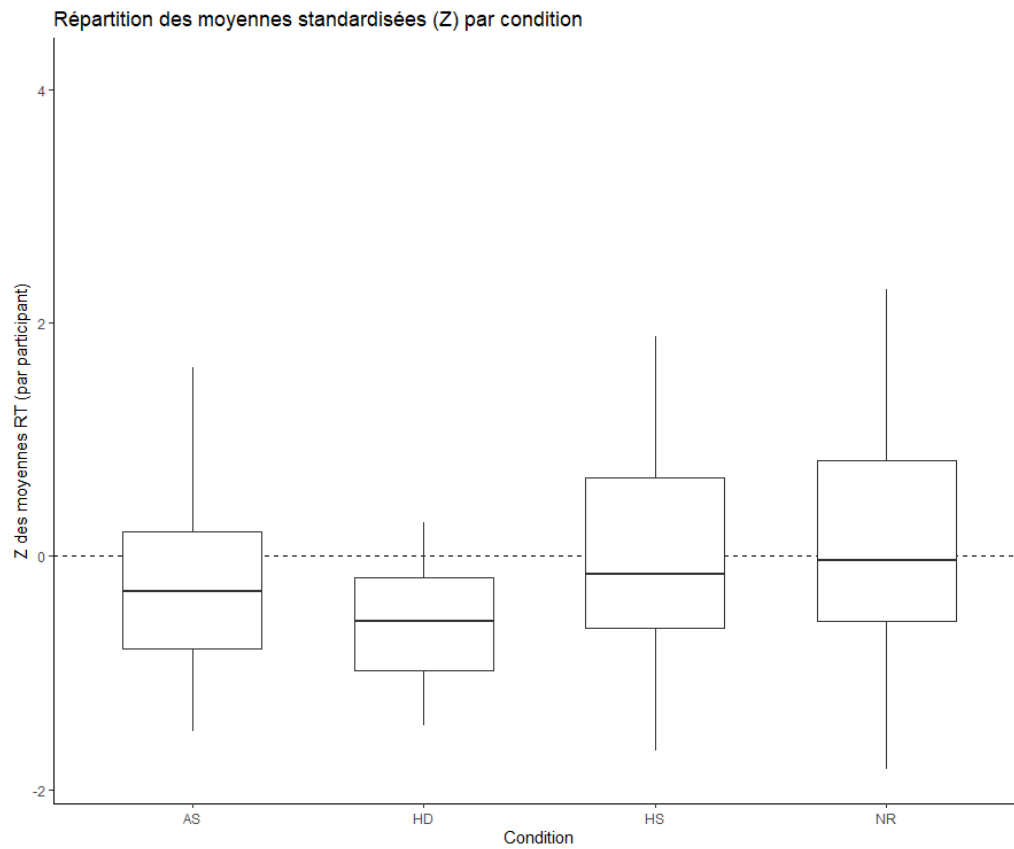


FIGURE 3 – F1 : distribution of means of RT for the individual prime-target pairs

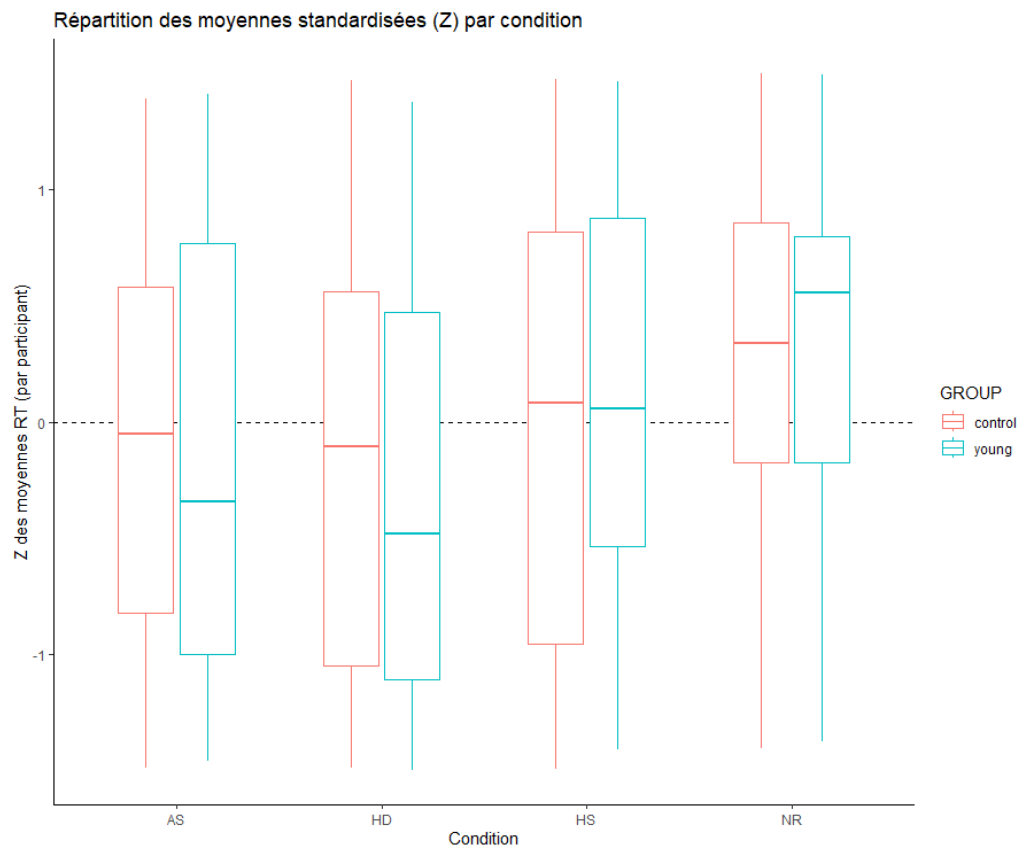


FIGURE 4 – F2 : distribution of means of RT for the individual participants among both groups

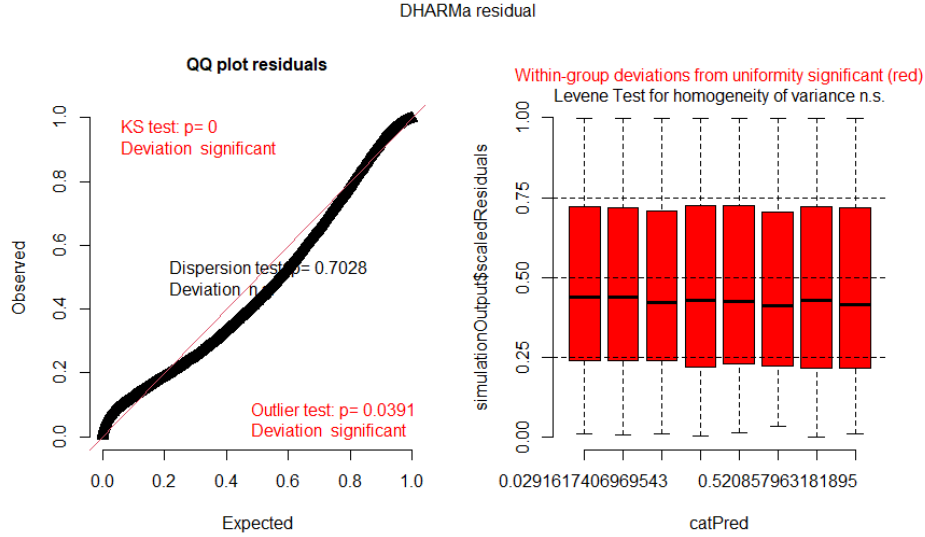


FIGURE 5 – Residuals of model m3

Final model retained was

```
m3 <- lmer(zscore condition + group + condition * group(1|paiofprimetarget), data = data, REML = TRUE)
```

As presented in Figure 5, residual DHARMA diagnostics indicated a significant deviation from uniformity in the KolmogorovSmirnov test (KS test, $p < .001$), as well as a marginally significant signal in the outlier test ($p = .039$). However, the dispersion test was not significant ($p = .70$), suggesting that there was no evidence of global heteroskedasticity. Visual inspection of the QQ-plot shows that the deviation is mainly located in the extreme tails of the distribution, while the central portion of the residuals aligns well with expectations. The boxplots by predictor level further indicate no systematic heterogeneity of variance across groups.

Taken together, these results suggest that the model captures the main structure of the data adequately, with only minor deviations from ideal assumptions that are unlikely to materially affect inference. Because there is no indication of strong heteroscedasticity or influential outliers, and the overall residual distribution is approximately symmetric and well-behaved, the model can reasonably be retained and interpretation of fixed effects can proceed.

4.1.2 Description of Model

Fixed effects : Fixed predictors were Condition and Group, plus their interaction. Condition had four levels (AD, AS, NR, and as reference level the M). Group had two levels (older vs. young).

Random effects : We included random intercepts for Prime-Target pairs ($n = 234$), allowing baseline Z-scores to vary across items. No random slopes were retained in the final model.

Factor levels and coding : Categorical factors used treatment (dummy) coding. **For Condition**, the baseline is conditionM, compared to conditionHD, conditionHS, conditionNR. **For Group**, the baseline is older adults; the coefficient GROUPyoung compares young to older

participants. Thus, the intercept estimates the mean z-score for the ConditionM in the older group, and each coefficient is interpreted as a deviation from that baseline.

Estimation and inference : Models were fitted by REML. Tests of fixed effects used Type-III ANOVA with Satterthwaite degrees of freedom.

4.1.3 control of influence of type of polysemy

According to the *Larousse* definition, *homonymy* occurs when two distinct words share the same form (spelling or pronunciation) but have completely unrelated meanings and often different origins. In contrast, *polysemy* refers to a single word that has several related meanings derived from a common origin or are linked by a shared idea. In our protocol, half of the words were polysemous and half were homonymous. In response to the reviewers concern, an additional analysis was conducted to test for a possible influence of this variable, but the effect was not significant. The control was done with the following code tests (done only on the ambiguous conditions) that seeked whether the variable **ambiguité** significantly improves model fit for zRT compared to a model without it. It did not.

```
dataamb <- subset(data, condition == "HD" | condition == "HS")

#full model
m_full <- lmer(z_score ~ condition + GROUP + ambiguité + condition*GROUP +
  (1 | prime_target), data = dataamb, REML = TRUE)

#reduced model
m_reduced <- lmer(z_score ~ condition + GROUP + ambiguité + condition*GROUP
  + (1 | prime_target), data = dataamb, REML = TRUE)

anova(m_full, m_reduced)

Data: dataamb
Models:
m_full: z_score ~ condition + GROUP + ambiguité + condition * GROUP + (1 |
  prime_target)
m_reduced: z_score ~ condition + GROUP + ambiguité + condition * GROUP + (1
  | prime_target)

```

	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr(>Chisq)
m_full	8	6544.2	6590.4	-3264.1	6528.2			
m_reduced	8	6544.2	6590.4	-3264.1	6528.2	0	0	

4.1.4 LDT : Model and ANOVA

```
> m3 <- lmer(z_score ~ condition + GROUP + condition*GROUP + (1 | prime_target),
data = data, REML = TRUE)
> summary(m3)
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: z_score ~ condition + GROUP + condition * GROUP + (1 | prime_target)
Data: data
```

REML criterion at convergence: 25714.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.2772	-0.7284	-0.1617	0.5743	3.4662

Random effects:

Groups	Name	Variance	Std.Dev.
prime_target	(Intercept)	0.1009	0.3176
Residual		0.8850	0.9407

Number of obs: 9313, groups: prime_target, 234

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-0.05817	0.05082	278.62139	-1.145	0.2533
conditionHD	-0.06593	0.08544	271.93094	-0.772	0.4410
conditionHS	0.09831	0.08520	280.65055	1.154	0.2495
conditionNR	0.12965	0.06123	280.40557	2.118	0.0351 *
GROUPyoung	-0.01693	0.04092	9077.25588	-0.414	0.6790
conditionHD:GROUPyoung	-0.00756	0.06793	9076.15222	-0.111	0.9114
conditionHS:GROUPyoung	0.04121	0.06885	9077.64174	0.598	0.5496
conditionNR:GROUPyoung	0.03626	0.04949	9080.26627	0.733	0.4638

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	cndtHD	cndtHS	cndtNR	GROUPy	CHD:GR	CHS:GR
conditionHD	-0.595						
conditionHS	-0.596	0.355					
conditionNR	-0.830	0.494	0.495				
GROUPyoung	-0.339	0.202	0.202	0.282			
cndHD:GROUP	0.204	-0.332	-0.122	-0.170	-0.602		
cndHS:GROUP	0.202	-0.120	-0.341	-0.167	-0.594	0.358	
cndNR:GROUP	0.281	-0.167	-0.167	-0.340	-0.827	0.498	0.491

```
> anova_m3 <- anova(m3, type = 3)
```

```
> effectsize(anova_m3)
```

Effect Size for ANOVA (Type III)

Parameter	Eta2 (partial)	95% CI
condition	0.06	[0.01, 1.00]
GROUP	6.15e-08	[0.00, 1.00]
condition:GROUP	1.07e-04	[0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].


```

-----
-----

> post_hoc1 <- pairs(emm,adjust = "FDR")
> df_posthoc <- as.data.frame(post_hoc1)
> df_posthoc$Cohen_d <- df_posthoc$estimate / sqrt(2) * df_posthoc$SE
> df_posthoc[, c("contrast", "estimate", "SE", "p.value", "Cohen_d")]
  contrast      estimate      SE    p.value    Cohen_d
1  M - HD  0.06970744 0.08077909 0.46689741  0.003981650
2  M - HS -0.11891627 0.08029052 0.20991961 -0.006751349
3  M - NR -0.14778491 0.05770216 0.03320436 -0.006029859*
4  HD - HS -0.18862371 0.09154697 0.08098187 -0.012210269
5  HD - NR -0.21749235 0.07255218 0.01813461 -0.011157823*
6  HS - NR -0.02886864 0.07200782 0.68885462 -0.001469911

```

4.2 CTTAT : Model and ANOVA

```
##### Accuracy Analysis 1 (selection)
#####

d_exp0$Type_c <- ifelse(d_exp0$condition1 == "global", +0.5, -0.5)
d_exp0$Distr_c <- ifelse(d_exp0$condition2 == "Distracteur R", +0.5, -0.5)
d_exp0$SetSize_c <- ifelse(d_exp0$numberdistractors == 3, +0.5, -0.5)
d_exp0$Group_c <- ifelse(d_exp0$GROUP == "young", +0.5, -0.5)

m_exp1_correct <- glmer(
  correct ~ Group_c * (Type_c + Distr_c + SetSize_c) +
    (1 + Type_c + Distr_c + SetSize_c | participant) +
    (1 | clue),
  data = d_exp0,
  family = binomial(link = "logit"),
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e5))
)

summary(m_exp1_correct)

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation)
['glmerMod']
Family: binomial ( logit )
Formula: correct ~ Group_c * (Type_c + Distr_c + SetSize_c) + (1 + Type_c +
Distr_c + SetSize_c | participant) + (1 | clue)
Data: d_exp0
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))

      AIC      BIC   logLik deviance df.resid
7651.0   7789.5  -3806.5   7613.0    10837

Scaled residuals:
      Min       1Q   Median       3Q      Max
-14.0165   0.1038   0.2118   0.3661   3.4504

Random effects:
Groups      Name      Variance Std.Dev. Corr
participant (Intercept) 0.45202  0.6723
              Type_c     0.73452  0.8570   0.08
              Distr_c    0.01385  0.1177   0.93 -0.01
              SetSize_c  0.72184  0.8496   0.23 -0.66  0.52
clue        (Intercept) 1.89405  1.3762
Number of obs: 10856, groups: participant, 184; clue, 59

Fixed effects:
              Estimate Std. Error z value      Pr(>|z|)
(Intercept)    2.3268    0.1972  11.800 < 0.0000000000000002 ***
Group_c         0.5539    0.1455   3.806    0.000141 ***
Type_c        -1.1221    0.3444  -3.259    0.001119 **
Distr_c        -0.1841    0.3359  -0.548    0.583504
SetSize_c      -0.3330    0.1515  -2.197    0.027990 *
Group_c:Type_c  -1.0788    0.1900  -5.677    0.0000000137 ***
Group_c:Distr_c  0.4601    0.1294   3.555    0.000378 ***
Group_c:SetSize_c -0.4021    0.2939  -1.368    0.171335
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
(Intr) Grop_c Type_c Dstr_c StSz_c Gr_:T_ Gr_:D_
Group_c      0.071
Type_c       0.060 -0.022
Distr_c      -0.014  0.004  0.009
SetSize_c    -0.114  0.071 -0.041 -0.001
Grp_c:Typ_c  -0.044 -0.088  0.066  0.009 -0.015
Grp_c:Dstr_   0.015  0.081 -0.005  0.045  0.013  0.027
Grp_c:StSz_   0.028 -0.252 -0.004 -0.002  0.154 -0.186  0.039
optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```
> table_means
```

```
# A tibble: 16 × 9
```

```
# Groups:   GROUP, condition1, condition2 [8]
```

GROUP	condition1	condition2	numberdistractors	mean_acc	sd_acc	n
se_acc	mean_acc_percent					
<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>
0.0143	1 control feature	Distracteur NR	1	0.91	0.287	400
0.00903	2 control feature	Distracteur NR	3	0.871	0.335	1376
0.0185	3 control feature	Distracteur R	1	0.838	0.369	400
0.0103	4 control feature	Distracteur R	3	0.825	0.380	1376
0.0239	5 control global	Distracteur NR	1	0.754	0.431	325
0.0132	6 control global	Distracteur NR	3	0.734	0.442	1118
0.0238	7 control global	Distracteur R	1	0.729	0.445	350
0.0132	8 control global	Distracteur R	3	0.699	0.459	1204
0.00796	9 young feature	Distracteur NR	1	0.971	0.168	447
0.00854	10 young feature	Distracteur NR	3	0.944	0.229	720
0.0118	11 young feature	Distracteur R	1	0.933	0.250	448
0.00939	12 young feature	Distracteur R	3	0.932	0.252	720
0.0220	13 young global	Distracteur NR	1	0.773	0.420	365
0.0197	14 young global	Distracteur NR	3	0.653	0.476	585
0.0204	15 young global	Distracteur R	1	0.796	0.404	392

16 young	global	Distracteur R	3	0.727	0.446	630
0.0178		72.7				


```
##### Accuracy Analysis 2 (controlled retrieval)
#####
```

```
d_exp2 <- subset(cttat_acc2, condition1 == "global")
```

```
lapply(d_exp2[, c("GROUP", "condition2", "condition3", "numberdistractors")],
function(x) unique(x))
```

```
## --- Contrasts centered  $\pm 0.5$  ---
```

```
d_exp2$Group_c <- ifelse(d_exp2$GROUP == "young", +0.5, -0.5)      # Group :
young = +0.5 ; control = -0.5
d_exp2$Dom_c <- ifelse(d_exp2$condition3 == "FORT", +0.5, -0.5)
d_exp2$Distr_c <- ifelse(d_exp2$condition2 == "Distracteur R", +0.5, -0.5)
d_exp2$SetSize_c <- ifelse(d_exp2$numberdistractors == 3, +0.5, -0.5)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace Approximation)
['glmerMod']
Family: binomial ( logit )
Formula: correct ~ Group_c * (Dom_c + Distr_c + SetSize_c) + (1 + Dom_c +
Distr_c + SetSize_c | participant) + (1 | clue)
Data: d_exp2
Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))
```

AIC	BIC	logLik	deviance	df.resid
4331.7	4455.4	-2146.8	4293.7	4950

Scaled residuals:

Min	1Q	Median	3Q	Max
-13.0237	-0.4715	0.2204	0.4483	3.0515

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
participant	(Intercept)	0.88529	0.9409	
	Dom_c	0.03158	0.1777	0.69
	Distr_c	0.02864	0.1692	-0.80 -0.12
	SetSize_c	0.01484	0.1218	-0.90 -0.94 0.45
clue	(Intercept)	2.74881	1.6580	

Number of obs: 4969, groups: participant, 184; clue, 38

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.89034	0.32817	5.760	0.0000000084 ***
Group_c	0.02845	0.18421	0.154	0.87727

```

Dom_c          -0.06794    0.52771  -0.129    0.89756
Distr_c        -0.02809    0.56473  -0.050    0.96033
SetSize_c      -0.52044    0.18654  -2.790    0.00527 **
Group_c:Dom_c   -0.00408    0.16434  -0.025    0.98019
Group_c:Distr_c  0.70169    0.16378   4.284  0.0000183278 ***
Group_c:SetSize_c -0.49545    0.36375  -1.362    0.17318

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

              (Intr) Grop_c Dom_c  Dstr_c StSz_c Grp_c:Dm_ Grp_c:Ds_
Group_c      0.002
Dom_c        -0.024 -0.008
Distr_c      -0.066  0.016 -0.101
SetSize_c    -0.135  0.113 -0.007 -0.011
Grop_c:Dm_c  0.012  0.125 -0.014  0.002  0.004
Grp_c:Dstr_  0.013 -0.097 -0.011 -0.005 -0.009  0.055
Grp_c:StSz_  0.042 -0.467  0.007 -0.014  0.046 -0.007    0.001

```

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

Accuracy Analysis 3 (color/size contrast)

#####

```

> d_exp2$Group_c <- ifelse(d_exp2$GROUP == "young", +0.5, -0.5)      # Group :
young = +0.5 ; control = -0.5

```

```

> d_exp2$feat_c <- ifelse(d_exp2$condition1 == "color", +0.5, -0.5)

```

```

> d_exp2$Distr_c <- ifelse(d_exp2$condition2 == "Distracteur R", +0.5, -0.5)

```

```

> d_exp2$SetSize_c <- ifelse(d_exp2$numberdistractors == 3, +0.5, -0.5)

```

```

> ##check no NA

```

```

> sapply(d_exp2[, c("Group_c", "feat_c", "Distr_c", "SetSize_c")], function(x)
any(is.na(x)))

```

```

  Group_c    feat_c    Distr_c SetSize_c
  FALSE     FALSE     FALSE     FALSE

```

```

> m_exp3_acc <- glmer(

```

```

+   correct ~ Group_c * (feat_c + Distr_c + SetSize_c) +

```

```

+   (1 + feat_c + Distr_c + SetSize_c | participant) +

```

```

+   (1 | clue),

```

```

+   data = d_exp2,

```

```

+   family = binomial,

```

```

+   control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e5))

```

```

+ )

```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation)
['glmerMod']

Family: binomial (logit)

Formula: correct ~ Group_c * (feat_c + Distr_c + SetSize_c) + (1 + feat_c +
Distr_c + SetSize_c | participant) + (1 | clue)

Data: d_exp2

Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))

AIC	BIC	logLik	deviance	df.resid
3336.4	3463.4	-1649.2	3298.4	5868

Scaled residuals:

Min	1Q	Median	3Q	Max
-9.0701	0.1277	0.1996	0.3090	2.9834

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
participant	(Intercept)	0.4734	0.6880	
	feat_c	0.5512	0.7424	-0.03
	Distr_c	0.2921	0.5404	0.74 0.09
	SetSize_c	1.1038	1.0506	0.63 -0.49 0.01
clue	(Intercept)	1.0366	1.0181	

Number of obs: 5887, groups: participant, 184; clue, 42

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.03977	0.20549	14.793	< 0.0000000000000002 ***
Group_c	1.11687	0.17574	6.355	0.000000000208 ***
feat_c	-0.20344	0.37540	-0.542	0.58788
Distr_c	-0.41775	0.37847	-1.104	0.26969
SetSize_c	-0.12237	0.18071	-0.677	0.49832
Group_c:feat_c	0.74284	0.24740	3.003	0.00268 **
Group_c:Distr_c	0.12371	0.23375	0.529	0.59662
Group_c:SetSize_c	-0.08398	0.34994	-0.240	0.81035

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Grop_c	feat_c	Dstr_c	StSz_c	Grp_:_	Gr_:D_
Group_c	0.133						
feat_c	-0.013	0.014					
Distr_c	-0.004	-0.009	0.013				
SetSize_c	-0.012	0.098	-0.020	-0.007			
Grop_c:ft_c	0.026	-0.012	0.130	0.009	-0.013		
Grp_c:Dstr_	0.002	0.078	0.005	0.137	0.021	0.076	
Grp_c:StSz_	0.044	-0.019	0.002	0.008	0.281	-0.099	0.031

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

RT Analysis 1 (selection)

#####

```

> cttat1 <- cttat1 %>%
+   mutate(`condition1` = case_when(
+     `condition1` == "Couleur" ~ "feature",
+     `condition1` == "taille" ~ "feature",
+     `condition1` == "synonyme" ~ "global",
+     TRUE ~ `condition1` # Conserve la valeur actuelle si aucune condition
n'est remplie
+   ))
> d_exp1 <- cttat1
> d_exp1$Type_c <- ifelse(d_exp1$condition1 == "global", +0.5, -0.5)
> d_exp1$Distr_c <- ifelse(d_exp1$condition2 == "Distracteur R", +0.5, -0.5)
> d_exp1$SetSize_c <- ifelse(d_exp1$numberdistractors == 3, +0.5, -0.5)
> d_exp1$Group_c <- ifelse(d_exp1$GROUP == "young", +0.5, -0.5)
> m_exp1_rt <- lmer(
+   z_score ~ Group_c * (Type_c + Distr_c + SetSize_c) +
+   (1 + Type_c + Distr_c + SetSize_c | participant) +
+   (1 | clue),
+   data = d_exp1,
+   control = lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e5))
+ )

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method

['lmerModLmerTest']

Formula: z_score ~ Group_c * (Type_c + Distr_c + SetSize_c) + (1 + Type_c + Distr_c + SetSize_c | participant) + (1 | clue)

Data: d_exp1

Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))

REML criterion at convergence: 21827.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.5630	-0.6998	-0.1917	0.5327	3.8835

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
participant	(Intercept)	0.001870285	0.043247	
	Type_c	0.221258447	0.470381	1.00
	Distr_c	0.004382483	0.066200	-0.34 -0.34
	SetSize_c	0.000006579	0.002565	0.97 0.97 -0.57
clue	(Intercept)	0.155191970	0.393944	
Residual		0.764017639	0.874081	

Number of obs: 8306, groups: participant, 183; clue, 59

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.075954	0.052800	57.099457	1.439	0.15575
Group_c	0.056868	0.022846	958.037723	2.489	0.01297 *
Type_c	0.052617	0.104080	94.568677	0.506	0.61436

Distr_c	0.250445	0.094927	83.436107	2.638	0.00994
**					
SetSize_c	0.020915	0.021429	7788.800678	0.976	0.32908
Group_c:Type_c	0.600117	0.081689	180.530690	7.346	0.000000000000678

Group_c:Distr_c	0.034038	0.040525	181.324654	0.840	0.40206
Group_c:SetSize_c	-0.008483	0.042776	7802.489495	-0.198	0.84282

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Grop_c	Type_c	Dstr_c	StSz_c	Gr_:T_	Gr_:D_
Group_c	0.010						
Type_c	0.102	0.026					
Distr_c	-0.016	-0.009	-0.006				
SetSize_c	-0.072	0.147	0.008	0.005			
Grp_c:Typ_c	0.016	0.342	0.066	0.000	0.006		
Grp_c:Dstr_	-0.007	-0.031	-0.012	0.013	-0.001	-0.093	
Grp_c:StSz_	0.033	-0.343	0.005	0.006	0.025	0.009	-0.007

optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

> table_means2

A tibble: 16 × 8

Groups: GROUP, condition1, condition2 [8]

	GROUP	condition1	condition2	numberdistractors	mean_rt	sd_rt	n	se_rt
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	control	feature	Distracteur NR	1	-0.112	0	346	0
2	control	feature	Distracteur NR	3	0.0590	0	1130	0
3	control	feature	Distracteur R	1	0.0323	0	314	0
4	control	feature	Distracteur R	3	0.339	0	1057	0
5	control	global	Distracteur NR	1	-0.0633	0	232	0
6	control	global	Distracteur NR	3	-0.287	0	788	0
7	control	global	Distracteur R	1	0.188	0	230	0
8	control	global	Distracteur R	3	-0.252	0	797	0
9	young	feature	Distracteur NR	1	-0.361	0	427	0
10	young	feature	Distracteur NR	3	-0.178	0	658	0
11	young	feature	Distracteur R	1	-0.0357	0	399	0
12	young	feature	Distracteur R	3	0.143	0	631	0
13	young	global	Distracteur NR	1	0.318	0	248	0
14	young	global	Distracteur NR	3	-0.0597	0	363	0
15	young	global	Distracteur R	1	0.314	0	285	0
16	young	global	Distracteur R	3	0.113	0	426	0

RT Analysis 2 (controlled retrieval)

#####

```
d_exp2 <- subset(cttat1, condition1 == "synonyme")

d_exp2$Dom_c      <- ifelse(d_exp2$condition3 == "fort", +0.5, -0.5)
d_exp2$Distr_c    <- ifelse(d_exp2$condition2 == "Distracteur R", +0.5, -0.5)
d_exp2$SetSize_c  <- ifelse(d_exp2$numberdistractors == 3, +0.5, -0.5)
d_exp2$Group_c    <- ifelse(d_exp2$GROUP == "young", +0.5, -0.5)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: z_score ~ Group_c * (Dom_c + Distr_c + SetSize_c) + (1 + Dom_c +
Distr_c + SetSize_c | participant) + (1 | clue)
Data: d_exp2
Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))
```

REML criterion at convergence: 8955.7

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.4872	-0.6820	-0.2060	0.5348	3.6742

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
participant	(Intercept)	0.0000000000026	0.000005099	
	Dom_c	0.088043276943	0.296720874	-1.00
	Distr_c	0.049550649070	0.222599751	-0.19 0.19
	SetSize_c	0.133568855735	0.365470732	-0.47 0.47 -0.21
clue	(Intercept)	0.242554816469	0.492498545	
Residual		0.777017006491	0.881485681	

Number of obs: 3355, groups: participant, 183; clue, 35

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.17172	0.09638	27.50223	1.782	0.0858 .
Group_c	0.30434	0.05462	78.60311	5.572	0.000000341 ***
Distr_c	0.14832	0.17826	31.25917	0.832	0.4117
SetSize_c	-0.26520	0.05451	78.07668	-4.866	0.000005815 ***
Group_c:Distr_c	-0.04209	0.07176	177.16985	-0.586	0.5583
Group_c:SetSize_c	0.05134	0.10890	77.73871	0.471	0.6386

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Grop_c	Dstr_c	StSz_c	Gr_:D_
Group_c	-0.005				
Distr_c	-0.056	0.003			
SetSize_c	-0.149	0.114	-0.018		
Grp_c:Dstr_	-0.012	-0.061	0.018	-0.009	
Grp_c:StSz_	0.044	-0.575	-0.010	0.019	-0.052

fit warnings:

fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

RT Analysis 3 (color/size contrast)

#####

```
d_exp2 <- subset(cttat3, condition1 == "color" | condition1 == "size")
```

```
unique(d_exp2$condition1)
```

```
d_exp2$feat_c <- ifelse(d_exp2$condition1 == "color", +0.5, -0.5)
```

```
d_exp2$Distr_c <- ifelse(d_exp2$condition2 == "Distracteur R", +0.5, -0.5)
```

```
d_exp2$SetSize_c <- ifelse(d_exp2$numberdistractors == 3, +0.5, -0.5)
```

```
d_exp2$Group_c <- ifelse(d_exp2$GROUP == "young", +0.5, -0.5)
```

```
## ===== Model LMM on zRT
```

```
m_exp2_rt <- lmer(
  z_score ~ Group_c * (feat_c + Distr_c + SetSize_c) +
    (1 + feat_c + Distr_c + SetSize_c | participant) +
    (1 | clue),
  data = d_exp2,
  control = lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e5))
)
```

```
summary(m_exp2_rt)
```

```
> summary(m_exp2_rt)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method

['lmerModLmerTest']

Formula: z_score ~ Group_c * (feat_c + Distr_c + SetSize_c) + (1 + feat_c + Distr_c + SetSize_c | participant) + (1 | clue)

Data: d_exp2

Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000))

REML criterion at convergence: 12823.6

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.3370	-0.6972	-0.1809	0.5446	3.6583

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
participant	(Intercept)	0.006693	0.08181	
	feat_c	0.145830	0.38188	-0.17
	Distr_c	0.010266	0.10132	0.87 -0.42
	SetSize_c	0.056102	0.23686	-0.70 0.08 -0.89
clue	(Intercept)	0.084136	0.29006	
Residual		0.726166	0.85215	

Number of obs: 4951, groups: participant, 183; clue, 39

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)	
(Intercept)	0.03549	0.05406	34.08348	0.657	0.515890	
Group_c	-0.21616	0.03727	80.13282	-5.801	0.000000126	***
feat_c	-0.38429	0.10313	43.41449	-3.726	0.000557	***
Distr_c	0.28668	0.09913	38.05615	2.892	0.006297	**
SetSize_c	0.19621	0.03736	80.14404	5.252	0.000001209	***
Group_c:feat_c	-0.32058	0.07593	178.89427	-4.222	0.000038475	***
Group_c:Distr_c	0.08863	0.05158	328.62453	1.718	0.086666	.
Group_c:SetSize_c	-0.02141	0.07468	79.97093	-0.287	0.775084	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Grop_c	feat_c	Dstr_c	StSz_c	Grp_:_	Gr_:D_
Group_c	-0.010						
feat_c	-0.010	-0.014					
Distr_c	0.014	0.002	-0.002				
SetSize_c	-0.197	0.118	0.016	-0.032			
Grop_c:ft_c	-0.010	-0.039	0.044	-0.010	0.008		
Grp_c:Dstr_	0.005	0.109	-0.009	0.018	-0.026	-0.106	
Grp_c:StSz_	0.046	-0.587	0.010	-0.003	-0.010	0.024	-0.134

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

4.3 Correlation matrix

```

dataacp <- data %>% dplyr::select (participant, z_score, condition, GROUP)
> dataacp2 <- cttat1 %>% dplyr::select (participant, z_score, condition1,
condition2, condition3, GROUP)
> dataacp2$z_score_sel1 <- dataacp2$z_score
> dataacp2$z_score_sel2 <- dataacp2$z_score
> dataacp2$z_score_ctrR <- dataacp2$z_score
> dataacp <- dataacp %>%
+   mutate(`condition` = case_when(
+     `condition` == "AS" ~ "prim_M",
+     `condition` == "HD" ~ "prim_AD",
+     `condition` == "HS" ~ "prim_AS",
+     TRUE ~ `condition` # Conserve la valeur actuelle si aucune condition
n'est remplie
+   ))
> tcd <- dataacp %>% ## index LDT
+   group_by(participant, condition, GROUP) %>% # Ajouter la colonne GROUP dans
le groupement
+   summarise(moyenne_z = mean(z_score, na.rm = TRUE)) %>%
+   pivot_wider(names_from = condition, values_from = moyenne_z)%>%
+   dplyr::ungroup()
`summarise()` has grouped output by 'participant', 'condition'. You can override
using the `.groups` argument.
> dataacp2 <- dataacp2 %>%
+   mutate(`condition1` = case_when(
+     `condition1` == "Couleur" ~ "feature",
+     `condition1` == "taille" ~ "feature",
+     `condition1` == "synonyme" ~ "global",
+     TRUE ~ `condition1` # Conserve la valeur actuelle si aucune condition
n'est remplie
+   ))
> tcd1 <- dataacp2 %>% ##index selection
+   group_by(participant, condition1) %>%
+   summarise(moyenne_z = mean(z_score_sel1, na.rm = TRUE)) %>%
+   pivot_wider(names_from = condition1, values_from = moyenne_z)%>%
+   dplyr::ungroup()
`summarise()` has grouped output by 'participant'. You can override using the
`.groups` argument.
> dataacp2 <- dataacp2 %>%
+   mutate(`condition2` = case_when(
+     `condition2` == "Distracteur R" ~ "rel.distr.",
+     `condition2` == "Distracteur NR" ~ "unr.distr",
+     TRUE ~ `condition2` # Conserve la valeur actuelle si aucune condition
n'est remplie
+   ))
> tcd2 <- dataacp2 %>% ##index selection
+   group_by(participant, condition2) %>%
+   summarise(moyenne_z = mean(z_score_sel2, na.rm = TRUE)) %>%
+   pivot_wider(names_from = condition2, values_from = moyenne_z)%>%
+   dplyr::ungroup()
`summarise()` has grouped output by 'participant'. You can override using the
`.groups` argument.
> d_exp <- subset(dataacp2, condition1 == "global")
> d_exp <- d_exp %>%
+   mutate(`condition3` = case_when(

```

```

+   `condition3` == "FAIBLE" ~ "ass_Sub.",
+   `condition3` == "FORT" ~ "ass_dom.",
+ ))
> tcd3 <- d_exp %>% ##index selection
+   group_by(participant, condition3) %>%
+   summarise(moyenne_z = mean(z_score_ctrR, na.rm = TRUE)) %>%
+   pivot_wider(names_from = condition3, values_from = moyenne_z)%>%
+   dplyr::ungroup()
`summarise()` has grouped output by 'participant'. You can override using the
`.groups` argument.
> data0ACP <- dplyr::inner_join(tcd, tcd1, by = "participant")
> data1ACP <- dplyr::inner_join(data0ACP, tcd2, by = "participant")
> dataACP <- dplyr::inner_join(data1ACP, tcd3, by = "participant")
> # ajouter stroop et fluences

```

```

df <- dataCorr %>%
+   dplyr::select("prim_AD",
+                 "prim_AS",
+                 "prim_M",
+                 "feature" ,
+                 "global",
+                 "rel.distr.",
+                 "unr.distr",
+                 "ass_Sub.",
+                 "ass_dom.",
+                 "Ph_F",
+                 "Sem_F",
+                 "StroopScore")

```

```

> res <- rcorr(as.matrix(df), type = "pearson")
> cor_mat <- res$r # matrice de corrélation
> p_mat <- res$P # matrice de p-values
> # Conversion en vecteur et ajustement
> p_adj <- p.adjust(p_mat, method = "BH") # méthode Benjamini-Hochberg (FDR)
> # Retour à la forme matricielle
> p_adj_mat <- matrix(p_adj, nrow = nrow(p_mat), ncol = ncol(p_mat),
+                      dimnames = dimnames(p_mat))
> p_adj_mat

```

	prim_AD	prim_AS	prim_M	feature	global	rel.distr.
unr.distr	ass_Sub.	ass_dom.				
prim_AD	NA	0.6908033788	0.0400190975	0.21027495	0.27225466	0.7501635
prim_AS	0.6908034	NA	0.0001427867	0.69745235	0.68399588	0.5542502
prim_M	0.0400191	0.0001427867	NA	0.68399588	0.69080338	0.5816900
feature	0.2102749	0.6974523467	0.6839958767	NA	0.00000000	0.7458317
global	0.2722547	0.6839958767	0.6908033788	0.00000000	NA	0.6908034
rel.distr.	0.7501635	0.5542502	0.5816900	0.7458317	0.6908034	NA

rel.distr.	0.7501635	0.5542502273	0.5816899708	0.74583171	0.69080338	NA
0.0000000	0.683995876669787	0.745831713949750				
unr.distr	0.7501635	0.5816899708	0.4658989279	0.71822310	0.68399588	0.0000000
NA	0.697748750178800	0.683995876669787				
ass_Sub.	0.2764758	0.9543939872	0.7501635128	0.00000000	0.00000000	0.6839959
0.6977488	NA	0.000000008577825				
ass_dom.	0.4658989	0.5130525515	0.6839958767	0.00000000	0.00000000	0.7458317
0.6839959	0.000000008577825	NA				
Ph_F	0.7458317	0.3931896230	0.6908033788	0.01406742	0.01347516	0.7501635
0.7501635	0.053910066763915	0.021816672818090				
Sem_F	0.7501635	0.6839958767	0.5816899708	0.02181667	0.02181667	0.1086171
0.1164607	0.122098913150157	0.029166549819697				
StroopScore	0.5816900	0.7501635128	0.4063103431	0.06601180	0.06601180	0.1614994
0.2382213	0.134725361096821	0.134725361096821				
	Ph_F	Sem_F	StroopScore			
prim_AD	0.745831714	0.750163513	0.58168997			
prim_AS	0.393189623	0.683995877	0.75016351			
prim_M	0.690803379	0.581689971	0.40631034			
feature	0.014067416	0.021816673	0.06601180			
global	0.013475160	0.021816673	0.06601180			
rel.distr.	0.750163513	0.108617136	0.16149938			
unr.distr	0.750163513	0.116460682	0.23822134			
ass_Sub.	0.053910067	0.122098913	0.13472536			
ass_dom.	0.021816673	0.029166550	0.13472536			
Ph_F	NA	0.002143538	0.84298013			
Sem_F	0.002143538	NA	0.05604593			
StroopScore	0.842980134	0.056045931	NA			

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